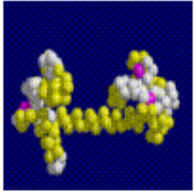


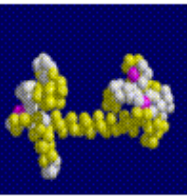
Applied bioinformatics: PSI-Blast



Morten Rye

Department of Clinical and
Molecular Medicine

*Slides from previous lectures by Prof. Finn
Drabløs*



MOL3000/MOL3100

□ Required reading

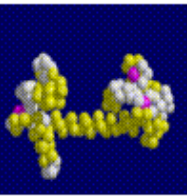
- Altschul SF and Koonin EV. Iterated profile searches with PSI-Blast – a tool for discovery in protein databases. Trends in Biochem. Sci. 23, 444-447 (1998)
- Jones DT and Swindells MB. Getting the most from PSI-Blast. Trends in Biochem. Sci. 27(93), 161-164 (2002)

□ Optional reading

- “Guide to BLAST home and search pages” (Under BLAST Help menu)
- Altschul SF *et al.* Gapped BLAST and PSI-Blast: a new generation of protein database search programs. Nucleic Acids Res. 25(17), 3389-3402 (1997)
- Johnson M *et al.* NCBI BLAST: a better web interface. Nucleic Acids Research 36, W5-W9 (2008)

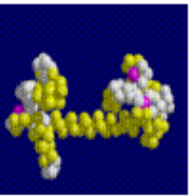
□ Main objective

- Learn about PSI-Blast and how to use it



The student shall be able to

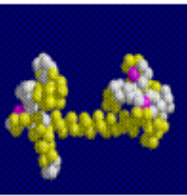
- ❑ describe the contents of typical sequence databases like SwissProt and Genbank / NCBI
- ❑ describe a standard sequence format like the Fasta format.
- ❑ explain how a dot plot is generated
- ❑ explain the general principle for pairwise sequence alignment
- ❑ describe the difference between pairwise and multiple alignment
- ❑ describe the output from ClustalX with respect to conserved and variable positions
- ❑ explain the difference between sequence and profile, and describe typical areas where this type of information is used
- ❑ explain the general principle for PSI-Blast, and explain the advantages of using PSI-Blast compared to Blast and similar pairwise methods
- ❑ explain how PSI-Blast can be used to find remote similarities, and in particular templates for homology-based modelling of 3D structures



Sequence alignment – Why?

Sequence alignment – How?

Application: Sequence Profiles
and PSI-BLAST



Will use PSI-Blast to look for similarities between sequences

□ DNA / RNA sequences

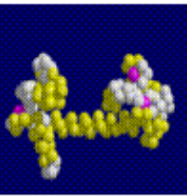
- 4 different bases

- ```
CACCGCTCCGGGCTCTCCGGTGGAGCCCGGTTGCTCGCCCGTGCCAGCCCCTACCCCGCG
GGGCCGCAGGGCAGAGGGCAACCCTGCTTTTCTGGGTAACTGGGCGAGCCTCATGCCTCTC
TGCCTTGCCTCTTCCCATGTGTAGCCTTTTGGGTTTGGGCAGCCGTTTGCAGCCAAAGC
TCTCGAATTCCGAAGAGGACGGACGGGTCTTTAAAGTACTCCATTCCAGGGCGGGCCTG
GGGATGCGTGCCCGGGATTCTTACCGATGGGCGCCTGGGG
```

## □ Protein sequences

- 20 different residues

- ```
MDKFLVRPDLRDLQGGGEEPAPTGGASGDLKSPDWRHLRAEGLSCDYTVLFGKAEADKIF
RELEQEVEYFTGALAKVQVFGKWHSVPRKQATYGDAGLTYTFSGLTLTTPKPWVPVLERVR
DRVCEVTGQTFNFVLVNRKYKDGCDHIGEHRDDERELAPGSPIASVSFGACRDFIFRHKDS
RGKRPRRTVEVVRLQLAHGSLMMNPPTNTHWYHSLPIRKRVLAPRVNLTFRKILPTKK
```



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[SNP](#)

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NCBI News & Blog

PubMed Labs is now part of NCBI Labs

03 Oct 2017

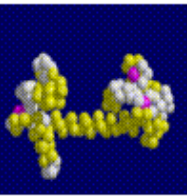
About two years ago, NCBI launched PubMed Labs, a gathering place for discovering and experimenting with new

October 11 NCBI Minute: Introducing the New RefSeq Functional Elements Project

02 Oct 2017

On October 11, 2017, NCBI will present a webinar on RefSeq Functional Elements

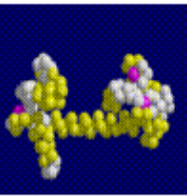
October 4th NCBI Minute: Create, link and share your bibliography (PubMed & ORCID)



□ Why are you comparing?

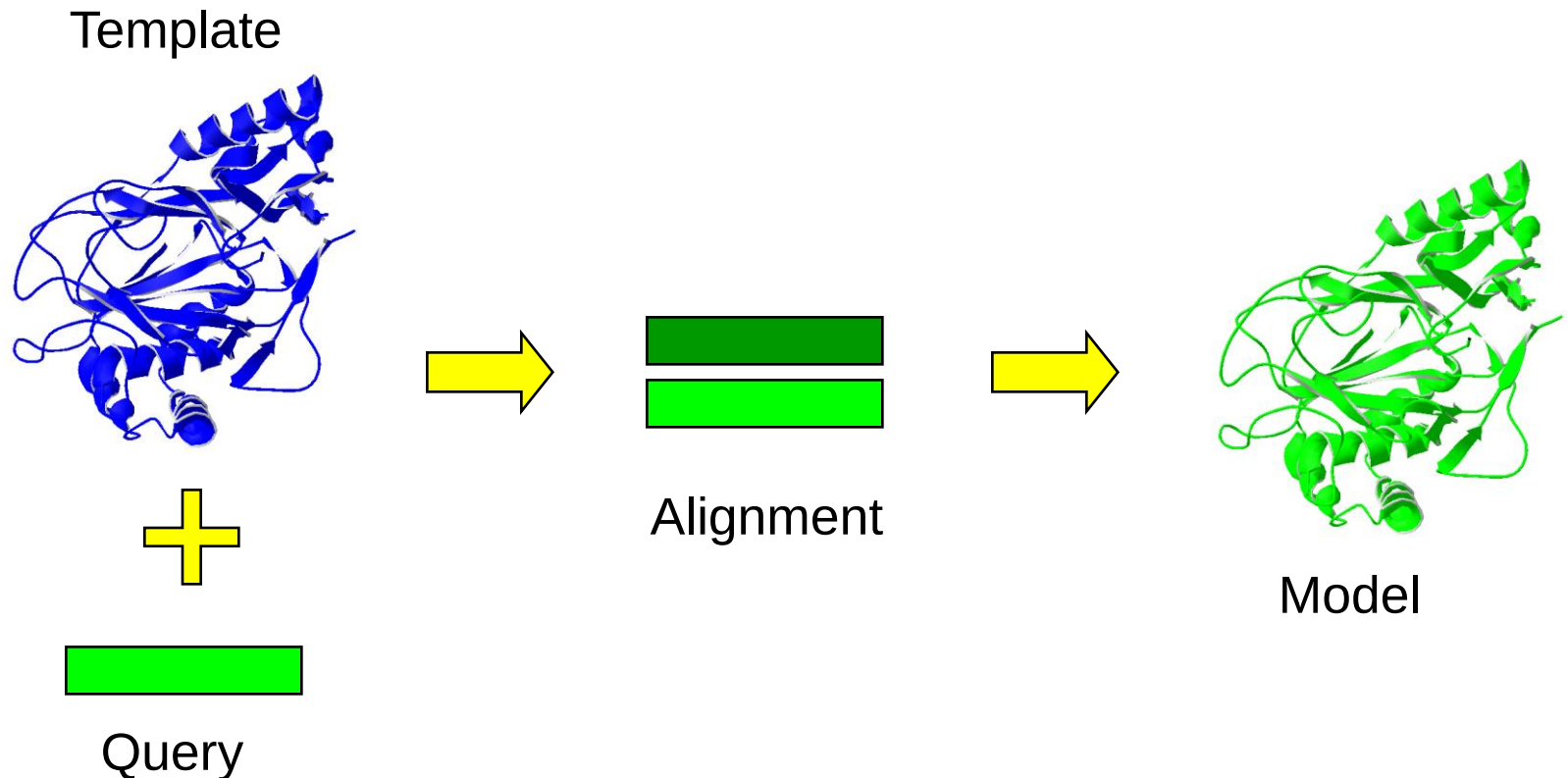
- Understand **evolution** of proteins - how they change over time
- Understand **function** by comparing sequences in proteins with similar function
- Understand the **differences** between similar proteins

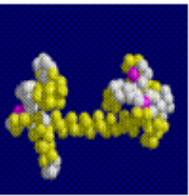
□ Application: **comparative model building**



Comparative model building

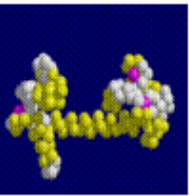
A fast approach to 3D data based on
homology (or similarity)



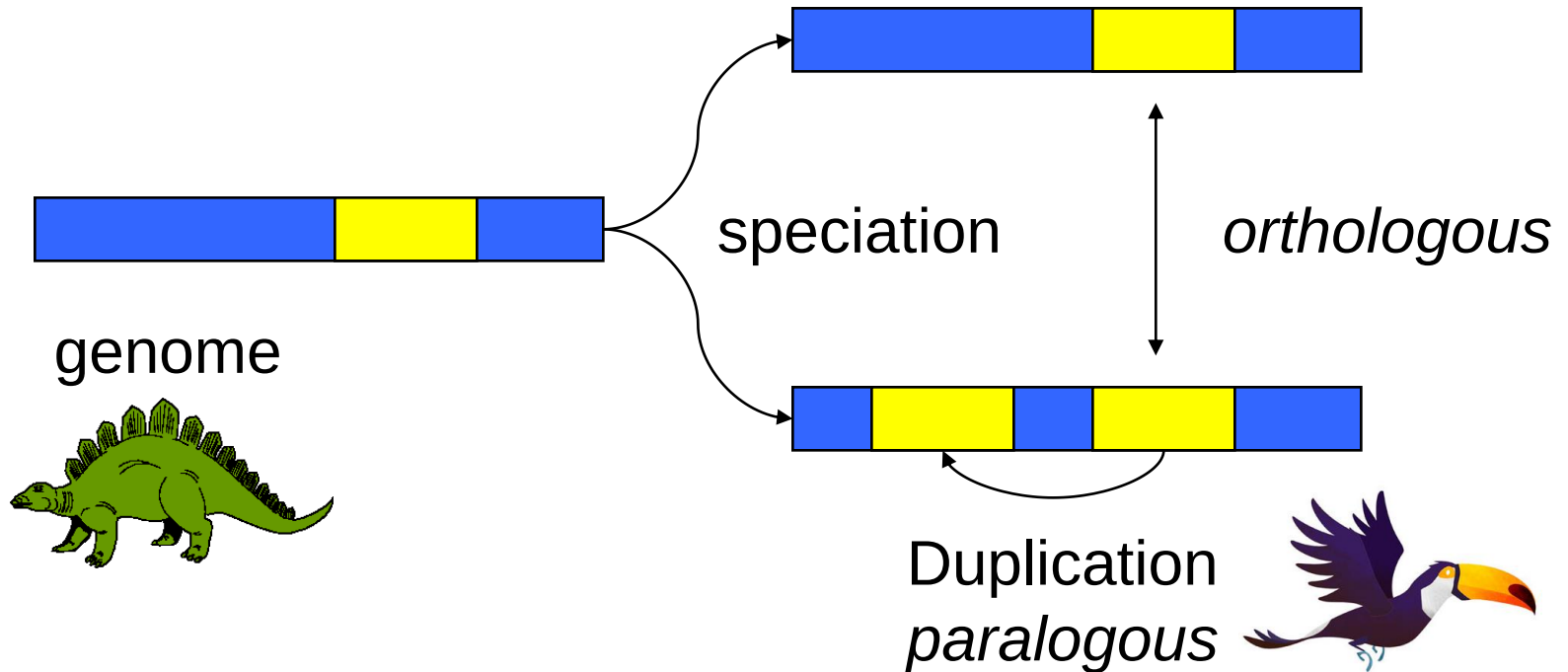


□ Expect similarity because:

- Evolutionary history: Proteins passed on from ancestors to new species will have similar **sequence**
- Protein with important function in one species: **Function** likely retained in evolution to new species
- But function can also **diverge**
- **Orthologs vs. paralog**s

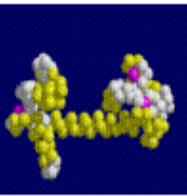


Orthologs vs. paralog



Orthologs are proteins from different species that evolved by vertical descent (speciation), and typically retain the same function as the original

Paralogs are proteins from within a given species that are derived from gene duplication, and may evolve new functions that are related to the original



Biochimica et Biophysica Acta 1804 (2010) 1695–1712



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Biochimica et Biophysica Acta

journal homepage: www.elsevier.com/locate/bbapap



Review

Computational approaches for protein function prediction: A combined strategy from multiple sequence alignment to molecular docking-based virtual screening

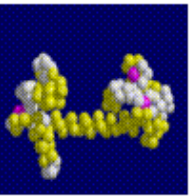
Ciro Leonardo Pierri ^{*}, Giovanni Parisi, Vito Porcelli

Department of Pharmaco-Biology, Laboratory of Biochemistry and Molecular Biology, University of Bari, Via E. Orabona, 4 - 70125 Bari, Italy

Orthologs, Paralogs, and Evolutionary Genomics¹

Eugene V. Koonin

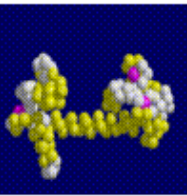
National Center for Biotechnology Information, National Library of Medicine,
National Institutes of Health, Bethesda, Maryland 20894;
email: koonin@ncbi.nlm.nih.gov



Sequence alignment – Why?

Sequence alignment – How?

Application: Sequence Profiles
and PSI-BLAST



Fasta/Pearson format

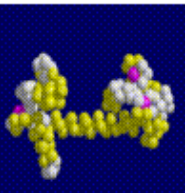
start

ID

text

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>1cdg_  mol:protein length:686      Cyclodextrin Glycosyltransferase (Cgtase)
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ISQPVENIYSIINYSGVNNTAYHGYWARDFKKTNPAYGTIADFQNLIAAAHAKNIKVIIDFAPNHTSPASSDQPS
FAENGRLYDNGTLLGGYTNDTQNLFHHNGGTDFTSTENGIYKNLYDLADLNHNNSTVDVYLKDAIKMWLDLGIDG
IRMDAVKHMPFGWQKSFMAAVNNYKPVFTFGEWFLGVNEVSPENHKFANESGMSLLDFRFAQKVRQVFRDNTDNM
YGLKAMLEGSAAADYAQVDDQVTFIDNHDMERFHASNANRRKLEQALAFITLTSRGVPAIYYGTEQYMSGGTDPDNR
ARIPSFSTSTTAYQVIQKLAPLRKCNPAIAYGSTQERWINNDVLIYERKFSGSNVAVVAVNRNLNAPASISGLVTS
LPQGSYNDVLGGLLNGNTLSVGSGGAASNFTLAAGGTAVWQYTAATATPTIGHVGPMMAKPGVTITIDGRGFGSS
KGTVYFGTTAVSGADITSWEDTQIKVKIPAVAGGNYNIAKANAAGTASNVYDNFEVLSGDQVSVRFVNNATTAL
GQNVYLTGSVSELGNWDPKAIGPMYNQVVYQYPNWYYDVSVPAGKTIEFKFLKKQGSTVTWEGGSNHTFTAPSS
GTATINVNWQP
```

no explicit end



SwissProt format

```
ID  AICDA_HUMAN      Reviewed;      198 AA.
AC  Q9GZX7; Q6QJ81; Q8NFC1;
DT  10-OCT-2002, integrated into UniProtKB/Swiss-Prot.
DT  01-MAR-2001, sequence version 1.
DT  18-SEP-2019, entry version 164.
DE  RecName: Full=Single-stranded DNA cytosine deaminase;
DE      EC=3.5.4.38 {ECO:0000269|PubMed:18722174};
DE  AltName: Full=Activation-induced cytidine deaminase;
DE      Short=AID;
DE  AltName: Full=Cytidine aminohydrolase;
GN  Name=AICDA; Synonyms=AID;
OS  Homo sapiens (Human).
OC  Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
OC  Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
OC  Catarrhini; Hominidae; Homo.
OX  NCBI_TaxID=9606;
RN  [1]
RP  NUCLEOTIDE SEQUENCE [GENOMIC DNA / MRNA] (ISOFORM 1).
RX  PubMed=10950930; DOI=10.1006/geno.2000.6268;
RA  Muto T., Muramatsu M., Taniwaki M., Kinoshita K., Honjo T.;
RT  "Isolation, tissue distribution, and chromosomal localization of the
RT  human activation-induced cytidine deaminase (hAID) gene.";
RL  Genomics 68:85-88(2000).
RN  [2]

(...)

FT  TURN      117      119      {ECO:0000244|PDB:5W0R}.
FT  HELIX      122      132      {ECO:0000244|PDB:5W0R}.
FT  STRAND     135      138      {ECO:0000244|PDB:5W0R}.
FT  HELIX      141      148      {ECO:0000244|PDB:5W0R}.
FT  HELIX      165      179      {ECO:0000244|PDB:5W0R}.
SQ  SEQUENCE   198 AA;  23954 MW;  3C27BB143DB184A9 CRC64;
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    FLRYISDWDL DPGRCYRVTW FTSWSPCYDC ARHVADFLRG NPNLSLRIFT ARLYFCEDRK
    AEPEGLRRLH RAGVQIAIMT FKDYFYCWNT FVENHERTFK AWEGLHENSF RLSRQLRRIL
    LPLYEVDDLK DAFRTLGL
```

//

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Entry

Publications

Feature viewer

Feature table

Protein | Single-stranded DNA cytosine deaminase

Gene | AICDA

Organism | *Homo sapiens* (Human)

Status |  Reviewed - Annotation score: ●●●●●● - Experimental evidence at protein levelⁱ

Functionⁱ

Single-stranded DNA-specific cytidine deaminase. Involved in somatic hypermutation (SHM), gene conversion, and class-switch recombination (CSR) in B-lymphocytes by deaminating C to U during transcription of Ig-variable (V) and Ig-switch (S) region DNA. Required for several crucial steps of B-cell terminal differentiation necessary for efficient antibody responses (PubMed:18722174, PubMed:21385873, PubMed:21518874, PubMed:27716525). May also play a role in the epigenetic regulation of gene expression by participating in DNA demethylation (PubMed:21496894). 5 Publications

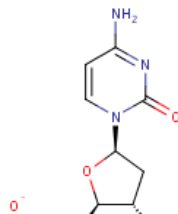
Catalytic activityⁱ

- a 2'-deoxycytidine in a single-stranded DNA + H⁺ + H₂O = a 2'-deoxyuridine in a single-stranded DNA + NH₄⁺ 1 Publication

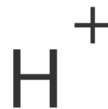
EC:3.5.4.38 1 Publication

Source: Rhea. [Hide](#)

a 2'-deoxycytidine in a single-stranded DNA



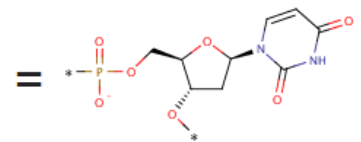
H⁺

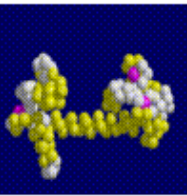


H₂O



a 2'-deoxyuridine in a single-stranded DNA





Sequence alignment

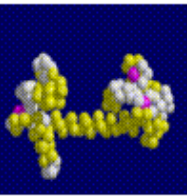
- ❑ Sequence alignments search for matches between sequences
- ❑ Two broad classes of sequence alignments
 - *Global*
 - *Local*
- ❑ Alignment can be performed between:
 - *Two sequences (pairwise alignment)*
 - *More sequences (multiple alignment)*

Q	K	E	S	G	P	S	S	S	Y	C
V	Q	Q	E	S	G	L	V	R	T	T

Global alignment

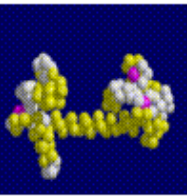
			E	S	G					
			E	S	G					

Local alignment



Biological importance of alignment

- Sequence alignments assess the degree of similarity between sequences
- Similar sequences suggest similar function
 - **Proteins** with similar sequences are likely to play similar biochemical roles
 - **Regulatory DNA sequences** that are similar will likely have similar roles in gene regulation
- Sequence similarity suggests evolutionary history
 - Fewer differences means more recent divergence (molecular clock)



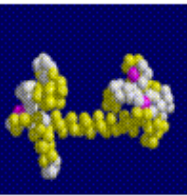
Algorithmic problems of alignments

- Comparison of similar sequences of similar length is straightforward
- How does one deal with insertions and gaps that may hide true similarity?
- How does one interpret minimal similarity?
 - Are sequences actually related?
 - Is alignment by chance?

Q	K	E	S	G	P	S	R	S	Y	C
Q	Q	E	S	G	P	V	R	S	T	C

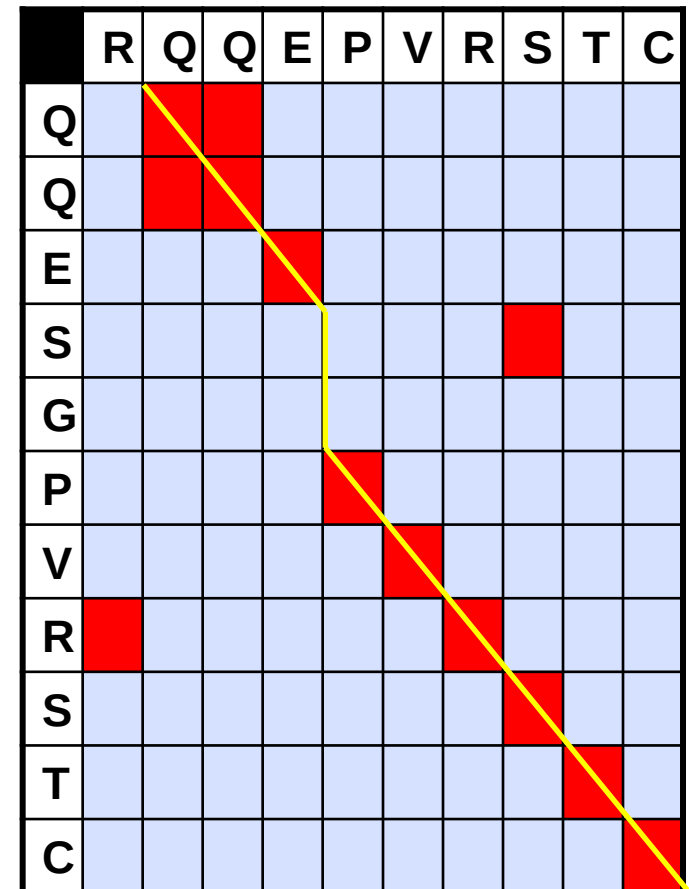
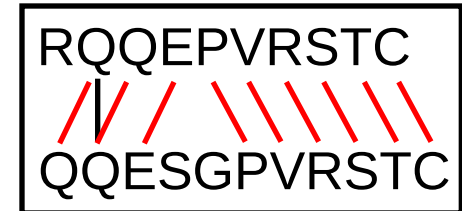
R	Q	Q	E	P	V	R	S	T	C	
/	/	/	/	/	/	/	/	/	/	
Q	Q	E	S	G	P	V	R	S	T	C

Q	K	G	S	Y	Q	E	K	G	Y	C
Q	Q	E	S	G	P	V	R	S	T	C



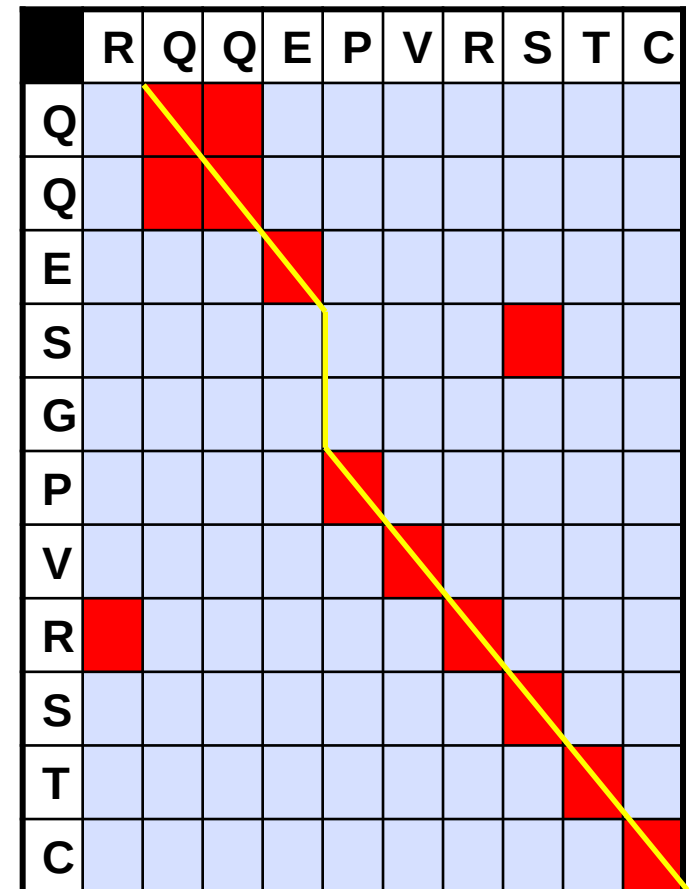
Gaps/Indels - Dot matrix analysis

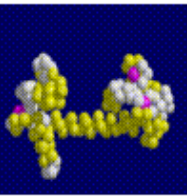
- A graphical method
- Shows all possible alignments
 - Red boxes show matches
 - String of red boxes indicate similarity
 - Yellow line show optimal alignment
 - Shift from diagonal indicate **indel** (insertion or deletion)





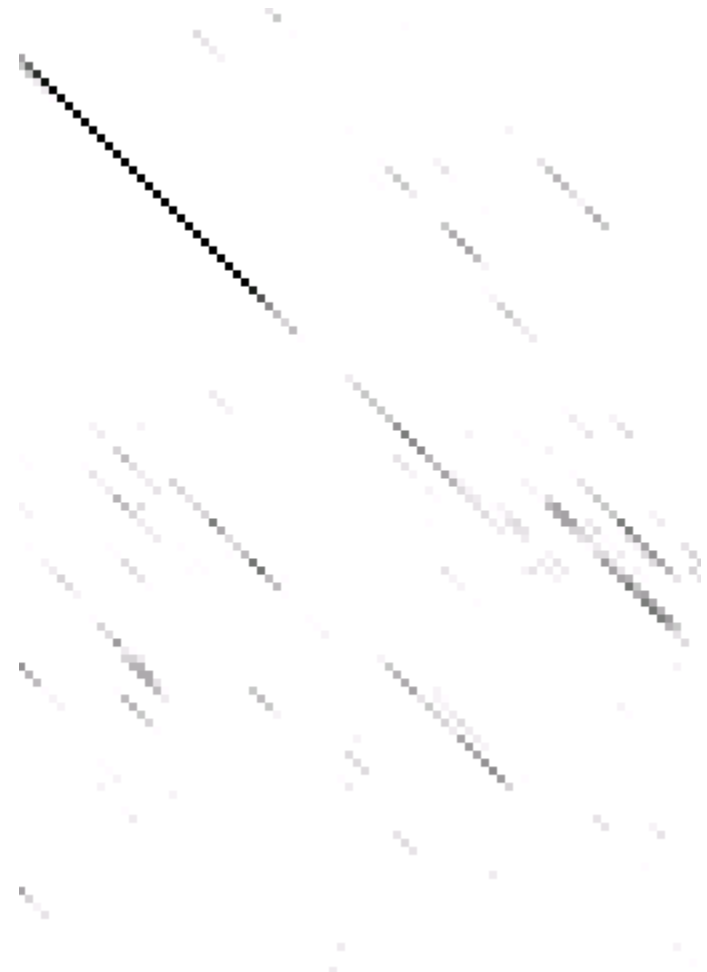
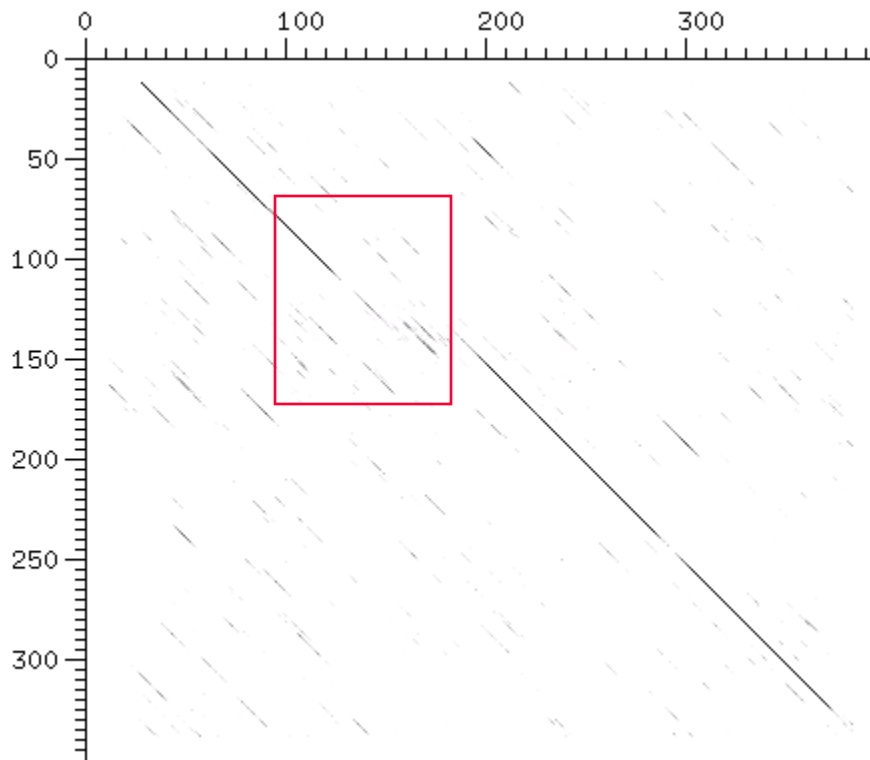
-
- RQQEPVIRSTC
 QQESGPVIRSTC

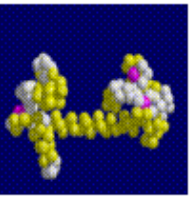




Dot matrix

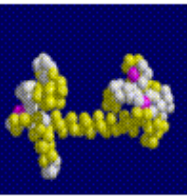
Protein-glutamate methyltransferase
CHEB_SALTY *Salmonella typhimurium*
CHEB_RHOCE *Rhodospirillum centenum*





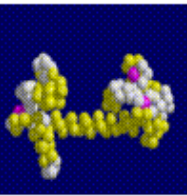
Optimal alignment

- Measure of similarity
 - *Mutation / scoring matrix - PAM, Blosum*
- Gap model
- Algorithm
 - *Dynamic Programming, HMM*



Scoring matrix

- Scoring matrices allow biologists to quantify the quality of sequence alignments
 - Score for: Amino acid A change (mutate) into amino acid B
- PAM and BLOSUM are two popular scoring matrices



An example of scoring

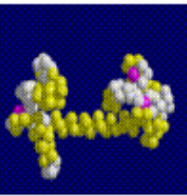
A sequence comparison

A	D	D	R	Q	C	E	R	A	D
A	Q	E	R	Q	E	C	Q	A	Q
4	0	2	5	5	-4	-4	1	4	0

Total score: 13

	A	R	N	D	C	Q	E
A	4	-1	-2	-2	0	-1	-1
R	-1	5	0	-2	-3	1	0
N	-2	0	6	1	-3	0	0
D	-2	-2	1	6	-3	0	2
C	0	-3	-3	-3	9	-3	-4
Q	-1	1	0	0	-3	5	2
E	-1	0	0	2	-4	2	5

BLOSUM62



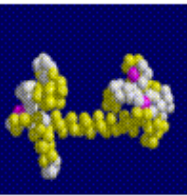
The scoring matrix problem

□ The problem

- A scoring matrix is normally based on statistics on how often residue type A is mutated to residue type B
- These statistics are obtained from real sequence alignments

□ How can we make these real alignments without a scoring matrix?

- Select very similar sequences (evolutionary closely related)
- Select sequences with very similar function (functional conservation - similar structural domains)



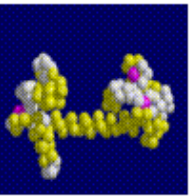
Making the scoring matrix

□ The PAM way

- Find closely related sequences and align these
 - Easy to align correctly, even by hand
 - ..., but evolutionary too close to be representative
- Assume initial model for mutation probabilities
- Extrapolate model to realistic evolutionary distances

□ The BLOSUM way

- Align functionally conserved regions in sequences
 - Can be aligned without gaps, easy to get correct
 - Can represent long evolutionary distances (no extrapolation needed)
- Count mutations for a given evolutionary distance
 - From degree of sequence variation



The PAM 250 scoring matrix

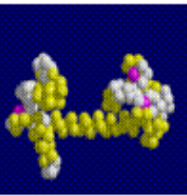
Point Accepted Mutation

1 PAM $\equiv$ 1 accepted substitution / 100 residues

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	...
A	2	-2	0	0	-2	0	0	1	-1	-1	-2	-1	-1	-3	1	1	1	-6	...
R	-2	6	0	-1	-4	1	-1	-3	2	-2	-3	3	0	-4	0	0	-1	2	...
N	0	0	2	2	-4	1	1	0	2	-2	-3	1	-2	-3	0	1	0	-4	...
D	0	-1	2	4	-5	2	3	1	1	-2	-4	0	-3	-6	-1	0	0	-7	...
C	-2	-4	-4	-5	12	-5	-5	-3	-3	-2	-6	-5	-5	-4	-3	0	-2	-8	...
Q	0	1	1	2	-5	4	2	-1	3	-2	-2	1	-1	-5	0	-1	-1	-5	...
E	0	-1	1	3	-5	2	4	0	1	-2	-3	0	-2	-5	-1	0	0	-7	...
G	1	-3	0	1	-3	-1	0	5	-2	-3	-4	-2	-3	-5	0	1	0	-7	...
H	-1	2	2	1	-3	3	1	-2	6	-2	-2	0	-2	-2	0	-1	-1	-3	...
I	-1	-2	-2	-2	-2	-2	-2	-3	-2	5	2	-2	2	1	-2	-1	0	-5	...
L	-2	-3	-3	-4	-6	-2	-3	-4	-2	2	6	-3	4	2	-3	-3	-2	-2	...
K	-1	3	1	0	-5	1	0	-2	0	-2	-3	5	0	-5	-1	0	0	-3	...
...																			

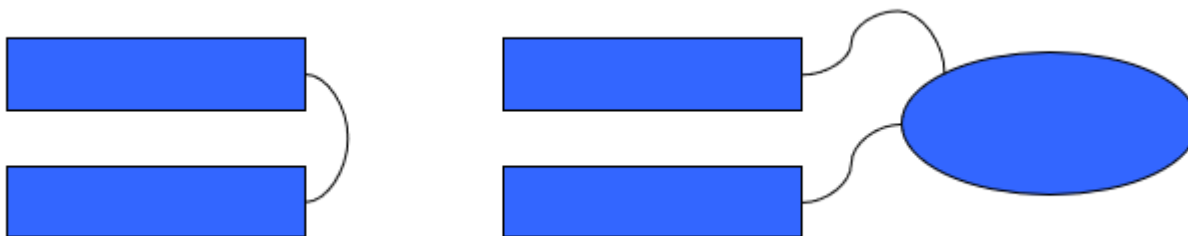


"pam" Version 1.0.6



Gap model

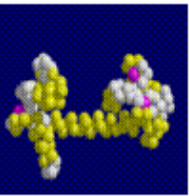
- Gap penalty
- Creating a new gap (opn – opening)
- Extending an existing gap (ext)
- Simple model - $GAP = ext \times len$
 - Only depend on length
- Improved model - $GAP = opn + ext \times (len - 1)$
 - Creating a new gap penalized more than extending an existing gap





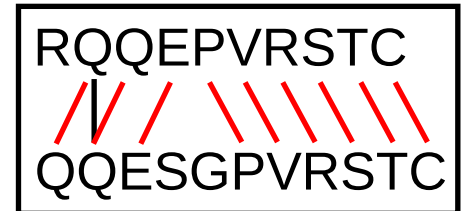
-
- RQQEPVIRSTC
 QQESGPVIRSTC

	R	Q	Q	E	P	V	R	S	T	C
Q										
Q										
E										
S										
G										
P										
V										
R										
S										
T										
C										

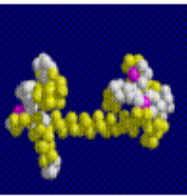


Dynamic Programming

- Breaks up the alignment into smaller (and simpler) subtasks
- Sequences are broken into smaller subsequences
 - *For example of length $k=3$*
- Each subsequence is scored
- Low scoring subsequences are removed
- **Decrease the number of sequence comparisons that is considered for the alignment (or optimal path)**
- ..but, still too slow for alignment to databases – needs shortcuts!

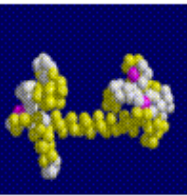


	R	Q	Q	E	P	V	R	S	T	C
Q										
Q										
E										
S										
G										
P										
V										
R										
S										
T										
C										



Fast alignment and searching

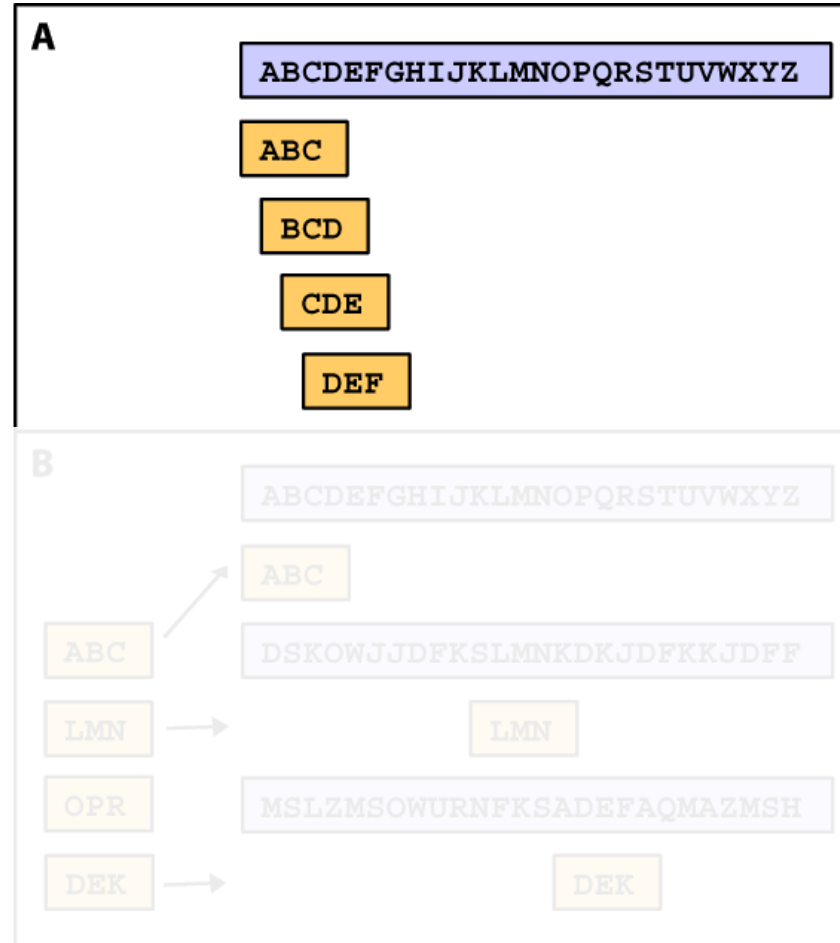
- Finds short identical matches
 - *A true match contains short high-scoring regions*
 - *Keeps track of short words (3 residues for proteins), searches for high-scoring hits*
- Search for diagonals with many matches
- Extends hits (initially without gaps)
- Banded search around diagonal
 - *Reduce search space*
- FASTA and Blast (we focus on Blast)

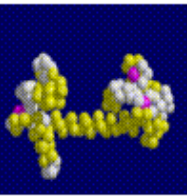


Example: Blast

From query sequence:

- derive a list of words (k -tuples) of length k (e.g., 3)
- Keep high-scoring words
 - *According to scoring matrix, i.e., the words that are most likely to represent conservation*

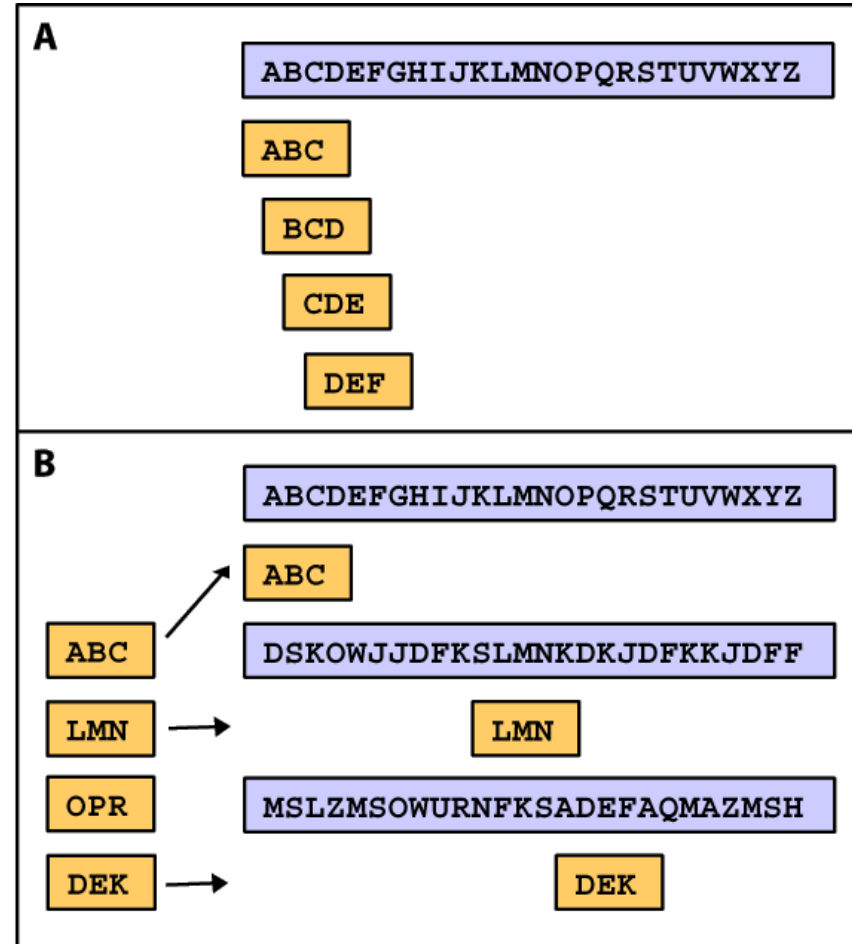


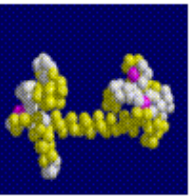


Example: Blast

□ Database sequences:

- High-scoring words are compared against each database sequence
- Sequences with many matches to high-scoring words are used for final alignments
- These alignments used as “anchors” to guide further alignments

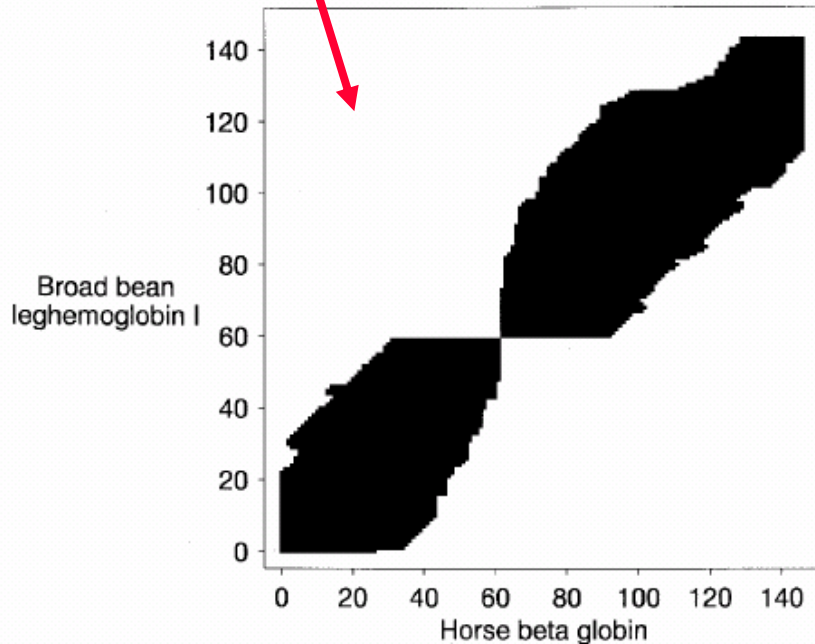




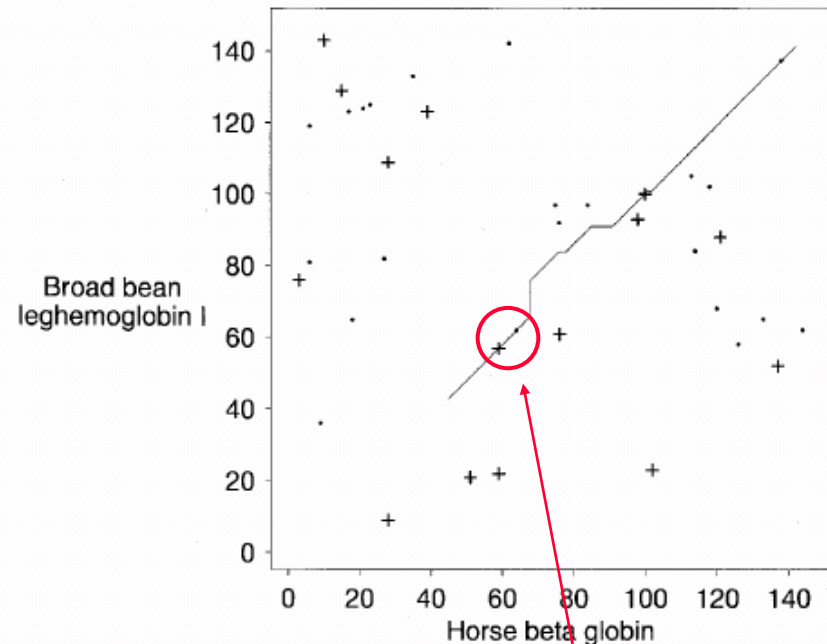
Banded search in matrix

Blast can ignore large parts of similarity matrix

a



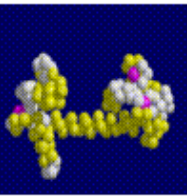
b



c

Leghemoglobin	43	FSFLKDSAGVVDSPKLGAAAEKVFGMVDRSAVQLRATGEVV--LDGKDGS-----	90
		F L + V+ +PK+ AH +KV L + GE V LD G+	
Beta globin	45	FGDLSNPGAVMGNPKVKAHGKKV-----LHSFGEGVHHLDNLKGTFALSE	90
Leghemoglobin	91	IHIQKGVLDP-HFVVVKEALLKTIKEASGDKWSEELSAAEVAYDGLATAI	140
		+H K +DP +F ++ L+ + G ++ EL A+++ G+A A+	
Beta globin	91	LHCDKLHVDPENFRLLGNVLVVVLARHFGKDFTPELQASYQKVVAGVANAL	141

High-scoring words and pairs



Blast output

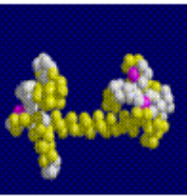
>SWALL:[6PGL SCHPO O74455](#) Probable 6-phosphogluconolactonase (EC 3.1.1.31) (6PGL).
Length = 257

Score = 336 (123.3 bits), Expect = 4.5e-30, P = 4.5e-30
Identities = 70/144 (48%), Positives = 96/144 (66%)

```
Query:    24 QVITINPEL----PVE--EAAEDYAKKLRQAFQGDS---IPVFDLLILGVGPDGHTCSLF 74
          ++ TINPEL      P++      A++Y K+L   F   S   +PVFDLL+LG GPDGHTCSLF
Sbjct:    99 KIHTINPELLKENPIDPQNVADEYEKQLVHVFANSSTVKVPVFDLLLLGCGPDGHTCSLF 158

Query:    75 PDHPLLQEREKIVAPISDSPKPPPQRVTLTLPVLNAARTVIFVATGEGKAAVLKRILEDQ 134
          PDH +LQE      VAP++DSPKPP  R+TLTLPV+  A+ + FV TG GK  +L  ++ED
Sbjct:   159 PDHEVLQEDVAWVAPVTDSPKPPKDRITLTLPVVTHAQAIAFVTTGAGKKDILPIVIEDF 218

Query:   135 EENPLPAALV-QPHTGKLCWFLDE 157
          LP+AL+ + + +  WF+D+
Sbjct:   219 TSK-LPSALITRNNLTRTSWFVDD 241
```



The Expect (E) value

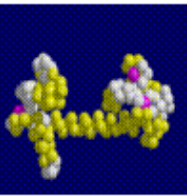
Score = 336 (123.3 bits), Expect = 4.5e-30, P = 4.5e-30
Identities = 70/144 (48%), Positives = 96/144 (66%)

□ Score

- The score of a specific alignment, based on values from the score matrix

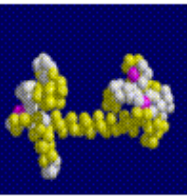
□ Expect

- The number of hits with similar or better score that one may expect to get just by chance (“false positive hits”)
- $E = 1$ – One should expect at least one random (non-significant) hit with similar or better score
- $E = 4.5 \times 10^{-30}$ – The likelihood of getting a hit with similar or better score is very small

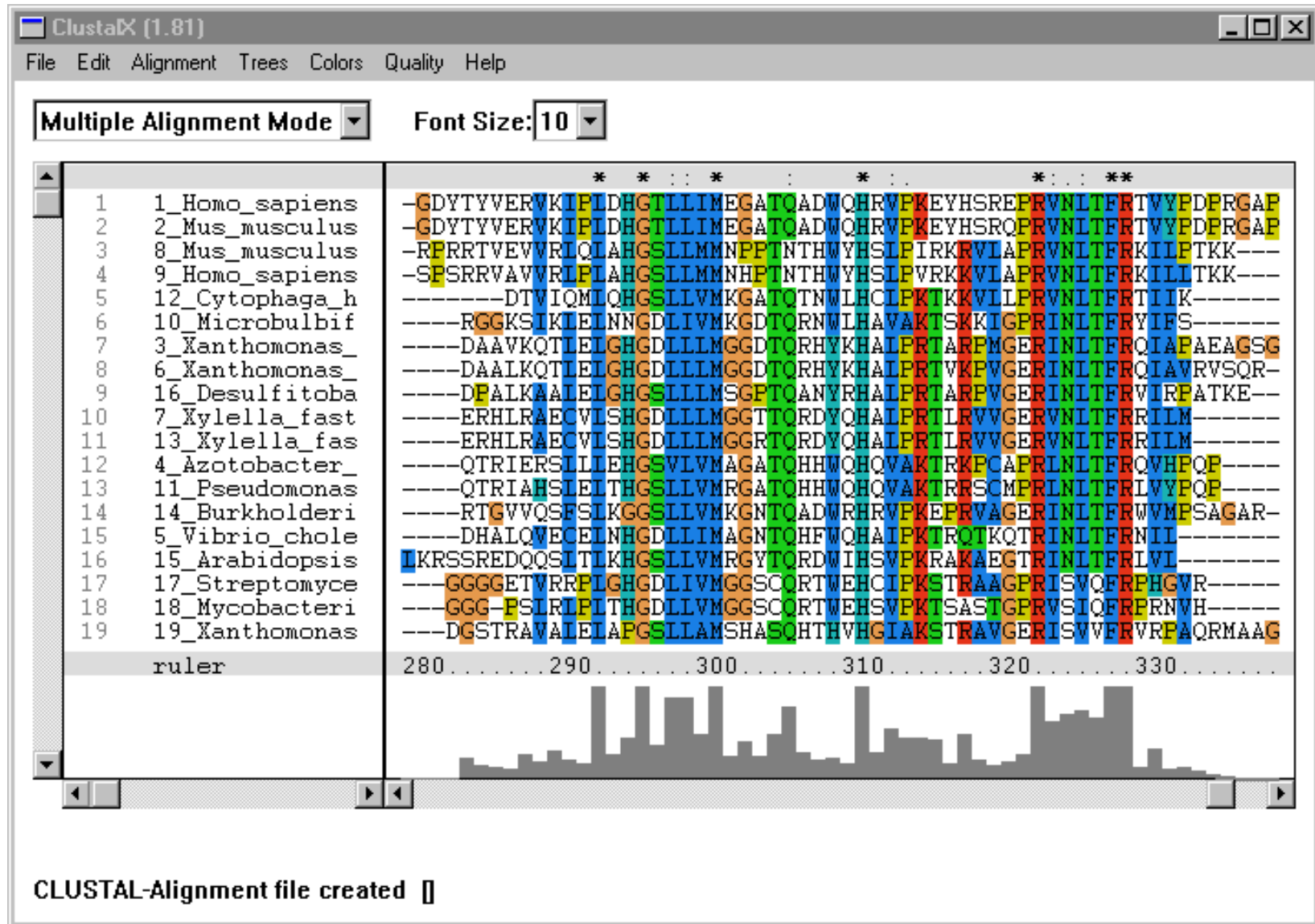


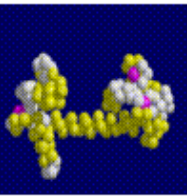
Hierarchical multiple alignment

- ❑ Multiple alignment
 - Find joint (shared) optimal alignment of several sequences
 - Resource demanding: Time grows with sequence length raised to number of sequences
- ❑ Hierarchical multiple alignment
 - Splits multiple alignment into several pairwise alignments
- ❑ Do pairwise alignment of all sequences
- ❑ Build “guide tree”
- ❑ Align sequences according to guide tree
 - Keep existing gaps
 - Insert new gaps when needed
 - No back-tracking for more optimal gaps ...
- ❑ Look carefully at alignment
 - In particular gaps
- ❑ Improve alignment with editor



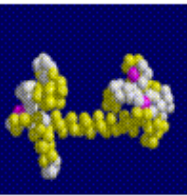
ClustalX (now Clustal Omega)





Conservation in ClustalX

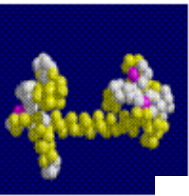
- “*” - Completely conserved (including no gaps ...)
- “:” - Conserved within one of the “strong” groups
 - STA, NEQK, NHQK, NDEQ, QHRK, MILV, MILF, HY, FYW
- “.” - Conserved within one of the “weak” groups
 - CSA, ATV, SAG, STNK, STPA, SGND, SNDEQK, NDEQHK, NEQHRK, FVLIM, HFY



Sequence alignment – Why?

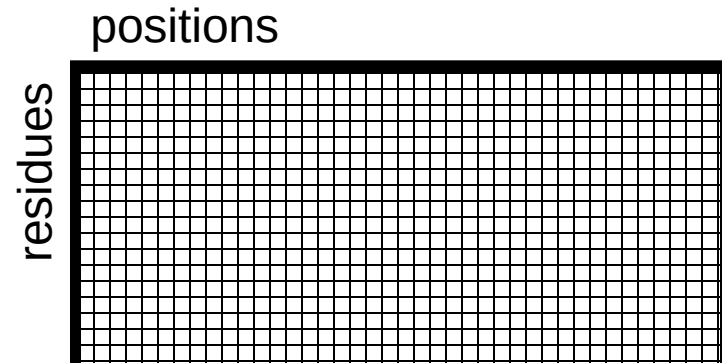
Sequence alignment – How?

**Application: Sequence
Profiles and PSI-BLAST**

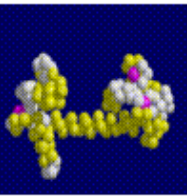


PSSMs / Profiles

	.									:	*	.	*	.	:	:	:	:	:	:																																		
HBB_HUMAN	F	S	D	G	L	A	H	L	D	N	-	-	-	-	L	K	G	T	F	A	T	L	S	E	L	H	C	D	K	L	H	V	D	P	E	N	F	R	L	L	G	N	V	L	V	C	V	L	A	H	H	E		
HBB_HORSE	F	G	E	G	V	H	H	L	D	N	-	-	-	-	L	K	G	T	F	A	A	L	S	E	L	H	C	D	K	L	H	V	D	P	E	N	F	R	L	L	G	N	V	L	V	V	L	A	R	H	E			
HBA_HUMAN	L	T	N	A	V	A	H	V	D	D	-	-	-	-	M	P	N	A	L	S	A	L	S	D	L	H	A	H	K	L	R	V	D	P	V	N	F	K	L	L	S	H	C	L	L	V	T	L	A	A	H	I		
HBA_HORSE	L	T	L	A	V	G	H	L	D	D	-	-	-	-	L	P	G	A	L	S	N	L	S	D	L	H	A	H	K	L	R	V	D	P	V	N	F	K	L	L	S	H	C	L	L	S	T	L	A	V	H	I		
GLB5_PETMA	V	N	D	A	V	A	S	M	D	D	T	-	-	-	E	K	M	S	M	K	L	R	D	L	S	G	K	H	A	K	S	F	Q	V	D	P	Q	Y	F	K	V	L	A	A	V	I	A	D	T	V	A	A	G	-
MYG_PHYCA	L	G	A	I	L	K	K	K	G	H	-	-	-	-	H	E	A	E	L	K	P	L	A	Q	S	H	A	T	K	H	K	I	P	I	K	Y	L	E	F	I	S	E	A	I	I	H	V	L	H	S	R	E		
LGB2_LUPLU	V	Y	E	A	A	I	Q	L	Q	V	T	G	V	V	T	D	A	T	L	K	N	L	G	S	V	H	V	S	K	G	-	V	A	D	A	H	F	P	V	V	K	E	A	I	L	K	T	I	K	E	V			



More focus on important positions / regions



A profile in DNA

Building a position-specific scoring matrix from a collection of sites

Alignment of Pho4p binding sites (TRANSFAC annotations)

R06098	T	C	A	C	A	C	G	T	G	G	G	A
R06099	G	G	C	C	A	C	G	T	G	C	A	G
R06100	T	G	A	C	A	C	G	T	G	G	G	T
R06102	C	A	G	C	A	C	G	T	G	G	G	G
R06103	T	T	C	C	A	C	G	T	G	C	G	A
R06104	A	C	G	C	A	C	G	T	T	G	G	T
R06097	C	A	G	C	A	C	G	T	T	T	T	C
R06101	T	A	C	C	A	C	G	T	T	T	T	C

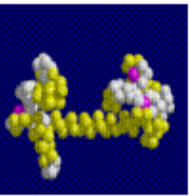
Count matrix (TRANSFAC matrix F\$PHO4_01)

Residue\position	1	2	3	4	5	6	7	8	9	10	11	12
A	1	3	2	0	8	0	0	0	0	0	1	2
C	2	2	3	8	0	8	0	0	0	2	0	2
G	1	2	3	0	0	0	8	0	5	4	5	2
T	4	1	0	0	0	0	0	8	3	2	2	2
Sum	8	8	8	8	8	8	8	8	8	8	8	8

Tom Schneider's sequence logo

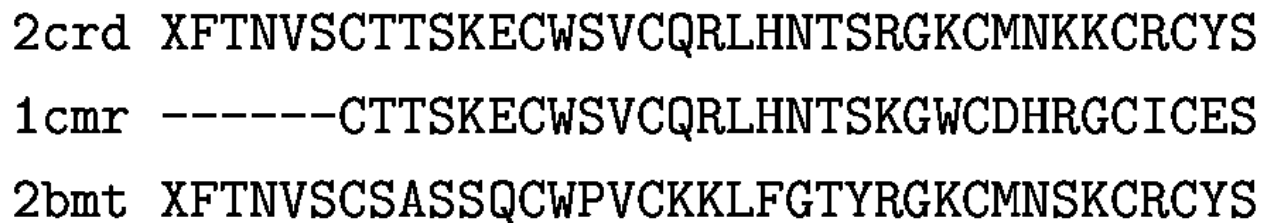
(generated with Web Logo <http://weblogo.barkley.cs.tulip.edu>)



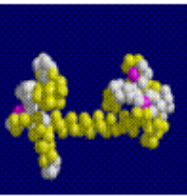


Iterative profile search (PSI-Blast)

1. Make an initial Blast search
2. Generate a multiple alignment of search results

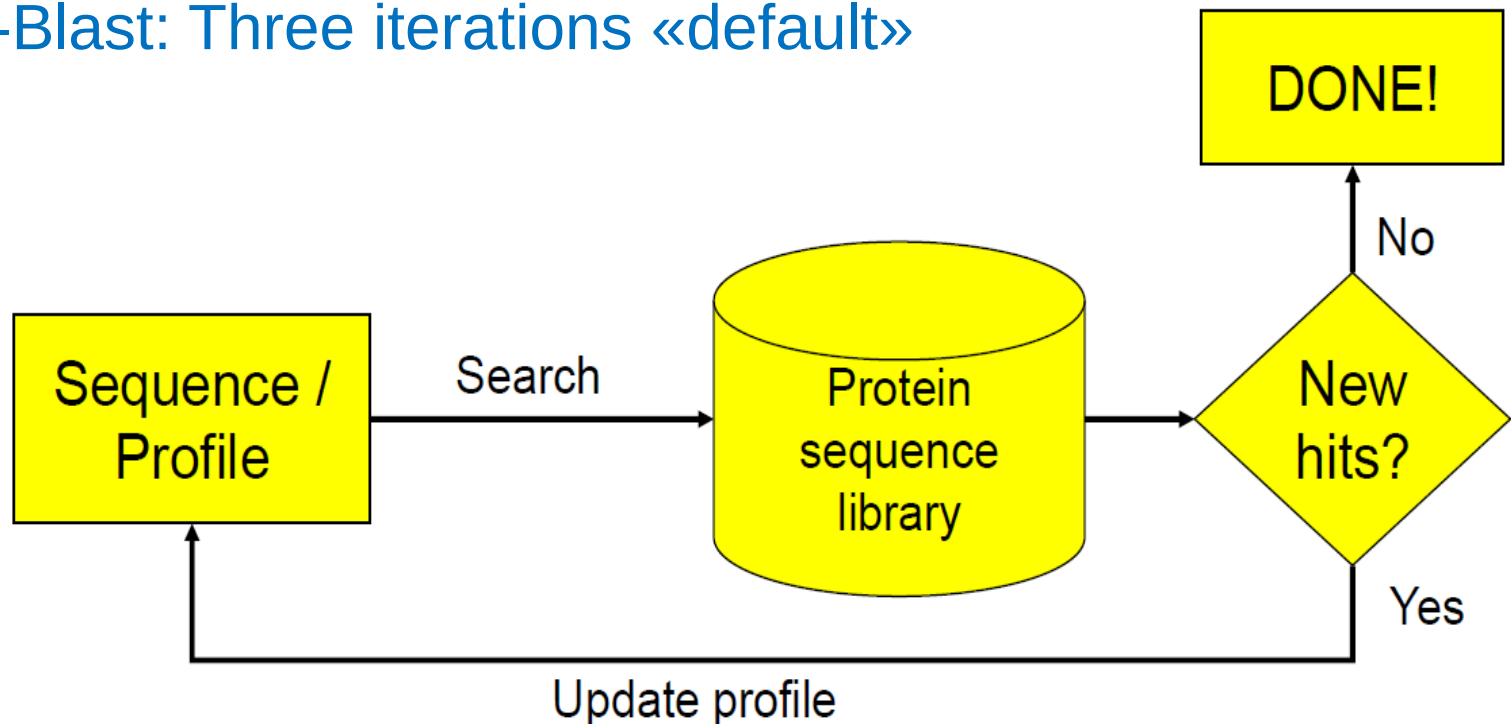


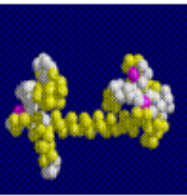
3. Create a PSSM from the multiple alignment
4. Perform a second iteration search with the PSSM / Profile)



Iterative profile search (PSI-Blast)

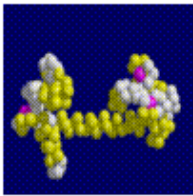
- Search more sensitive to conserved (important) positions
- Update profile with new results
- Perform next iteration etc...
- Continue until no new hits or satisfactory results
- PSI-Blast: Three iterations «default»





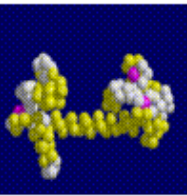
Evaluating results - some issues to be aware of

- ❑ Results are biased by sequence composition
 - High complexity sequences give robust results
 - Filter out low complexity regions / repeats (automatically)
 - Consider transmembrane regions, coiled coil regions (low to intermediate complexity)
- ❑ Significance depends on database size
 - Be careful when comparing different searches
- ❑ Iterating can also enhance errors!
- ❑ Remember: "Bioinformatic tools are predictive and should be used with appropriate care and attention."



An example

- Warning: Your output will be different, if you try to run this particular example. That is because of growing databases, new software versions etc.



Get the sequence (old)

NCBI Resources How To

Protein Protein NP_247388
Advanced

Display Settings: GenPept Send to:

hypothetical protein MJ_0414 [Methanocaldococcus jannaschii DSM 2661]
NCBI Reference Sequence: NP_247388.1
[FASTA](#) [Graphics](#)

Go to:

LOCUS NP_247388.1
DEFINITION hypothetical protein MJ_0414 [Methanocaldococcus jannaschii DSM 2661]
ACCESSION NP_247388
VERSION NP_247388.1
DBLINK Project: NCBI
DBSOURCE RefSeq
KEYWORDS RefSeq
SOURCE Methanocaldococcus jannaschii
ORGANISM Methanocaldococcus jannaschii
REFERENCE 1 (r)

NCBI Resources How To

Protein Protein
Advanced

Display Settings: FASTA

hypothetical protein MJ_0414 [Methanocaldococcus jannaschii DSM 2661]
NCBI Reference Sequence: NP_247388.1
[GenPept](#) [Graphics](#)

Example from
original PSI-
Blast paper

```
>gi|15668590|ref|NP_247388.1| hypothetical protein MJ_0414 [Methanocaldococcus  
jannaschii DSM 2661]  
MRLIMMKVSAYDLNKAIEKLNLSIKDLNKAFSRKILREDEYKEIKTLLFKKEFKGIEKGTVIFLNDNLDV  
VRGYPKTYRAITLYPTIKKHFDKVVIEEKLNGYNIRIVKIDGELYALTRSGYICPFTTKKVKKFLNLEI  
LDDYSEYMLCGEMIGINNPYTPYYYKEVDRGFENLGFYIFDIKERETNKSPLIKERINLCEKYNLPYVKP  
LAVVDKDEAHIHVREIIIEKLNKEGREGVVLKDPDMAVSPIKYTTHTYQCEDLKSFTFFFDLGMDFLFSR  
VVREGFMSYEFKETLEERKNRAKDLGEAILLPMVETINKVASGERVSEDFELIFDSEEDFDEFDLMRKM  
KMVITIKNIEKIDTEEGVKIKAVIGKIYNKTNDKIISYLNGLTWE
```



Same sequence
as original PSI-
Blast paper

 This record is a non-redundant protein sequence. Please [read more here](#).

NCBI Resources ☒ How To ☒

Protein

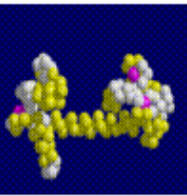
```

LOCUS       WP_010869913
DEFINITION  RNA ligase [Methanocaldococcus]
ACCESSION   WP_010869913
VERSION     WP_010869913.1
KEYWORDS    RefSeq.
SOURCE      Methanocaldococcus
   ORGANISM Methanocaldococcus
             Archaea; Euryarchaeota;
             Methanocaldococcaceae
COMMENT     REFSEQ: This record represents a
             sequence which may be identical
             from the same, or different,
             COMPLETENESS: full length.
FEATURES             Location/Qualifiers
     source           1..395
                     /organism="Methanocaldococcus"
                     /db_xref="taxon:10924"
     Protein          1..395

```

 This record is a non-redundant protein sequence. Please [read more here](#).

>WP_010869913.1 RNA ligase [Methanocaldococcus jannaschii]
MRLIMMKVSAYDLNKIAEKLNLISIKDLNKAFSRKILREDEYKEIKTLLFKKEFKGIEKGTIVFLNDNLDV
VRGYPKTYRAITLYPTIKKHFIDKVVIEEKLNGYNIRIVKIDGEVYALTRSGYICPFTTKVKKKFLNLEI
LDDYSEYMLCGEMIGINNPYTPYYYKEVDRGFENLGFYIFDIKERETNKSPLIKERINLCEKYNLPYVKP
LAVVDKDEAHIHVREIIIEKLNKEGREGVVLKDPDMAVSPIKYTTHYTQCEDLKSFTFFFDLGMDFLFSR
VVREGFMSYEFKETLEERKNRAKDLGEAILLPMVETINKVASGERVSEDFELIFDSEEDFDFLDFMRKM
KMITIKNIEKIDTEEGVKIKAVIGIKIYNKTNDKIISYLNGLTWE



Go to Protein Blast

 U.S. National Library of Medicine

NCBI National Center for Biotechnology Information

Sign in to NCBI

BLAST®

HomeRecent ResultsSaved StrategiesHelp

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

NEWS

Magic-BLAST 1.3.0 released
A new version of the BLAST RNA-seq mapping tool is now available.
Thu, 28 Sep 2017 16:00:00 EST
[More BLAST news...](#)

Web BLAST

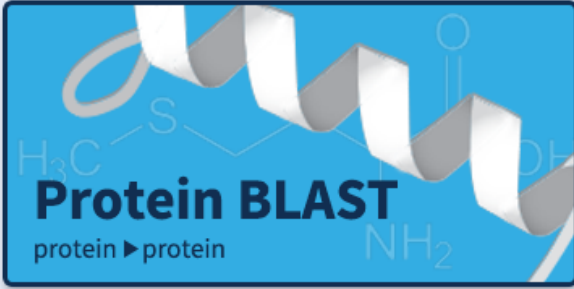


Nucleotide BLAST

nucleotide ► nucleotide

blastx
translated nucleotide ► protein

tblastn
protein ► translated nucleotide



Protein BLAST

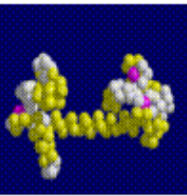
protein ► protein

BLAST Genomes

Search

HumanMouseRatMicrobes

<http://blast.ncbi.nlm.nih.gov/Blast.cgi>



PSI-Blast

blastn blastp **blastx** tblastn tblastx

BLASTP programs search protein databases using a protein query. [more...](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#) Query subrange [?](#)

From

To

Or, upload file No file selected. [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database [?](#)

Organism [Optional](#) ☐ exclude

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Exclude [Optional](#) ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)

☐ blastp (protein-protein BLAST)

☒ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

BLAST Search database nr using PSI-BLAST (Position-Specific Iterated BLAST)

☐ Show results in a new window

[+ Algorithm parameters](#)

sequence (fasta)
sequence range

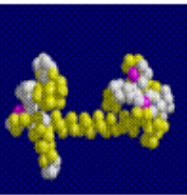
job title

database

limits

Blast tool

several advanced
options



Start iterations

Submit

Wait ...

Get the result!

[← Edit Search](#) [Save Search](#) [Search Summary](#) ▼

Job Title WP_010869913.1 RNA ligase [Methanocaldococcus...]

RID [T3R4AYP2014](#) Search expires on 10-01 18:44 pm [Download All](#) ▼

Program PSI-BLAST Iteration 1 [Citation](#) ▼

Database **pdb** [See details](#) ▼

Query ID [tblQuery_290060](#)

Description WP_010869913.1 RNA ligase [Methanocaldococcus jannaschii]

Molecule type amino acid

Query Length 395

Other reports [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#) ?

[? How to read this report?](#) [BLAST Help Videos](#) [Back to Traditional Results Page](#)

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

[+ Add organism](#)

Percent Identity to **E value** to **PSI-BLAST incl. threshold**

[Filter](#) [Reset](#)

Run PSI-Blast iteration 2

Number of sequences [Run](#)

Descriptions

[Graphic Summary](#)

[Alignments](#)

[Taxonomy](#)

Sequences producing significant alignments

[Download](#) ▼ [Manage Columns](#) ▼ Show ?

18 sequences selected

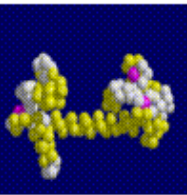
[GenPept](#) [Graphics](#) [Distance tree of results](#) [Multiple alignment](#)

Sequences with E-value BETTER than threshold

☒ select all 4 sequences selected

PSI-BLAST iteration 1

	Description	Max score	Total score	Query cover	E value	Per. Ident	Accession	Select for PSI blast	Used to build PSSM	Newly added
☒	Chain A. Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]	313	313	95%	4e-104	42.26%	5D1O_A	☒		
☒	Chain B. Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]	311	311	95%	3e-103	41.99%	5D1O_B	☒		
☒	Chain A. PAB1020 [Pyrococcus abyssi GE5]	183	183	90%	2e-53	32.96%	2VUG_A	☒		
☒	Chain A. DNA ligase [Aquifex aeolicus]	158	158	85%	2e-44	30.59%	3QWU_A	☒		



Iteration 1

Hits are listed and shown as alignments

[Download](#) [GenPept](#) [Graphics](#)

Chain A, Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]
Sequence ID: [5D10_A](#) Length: 378 Number of Matches: 1

Range 1: 10 to 378 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
313 bits(801)	4e-104	Compositional matrix adjust.	161/381(42%)	234/381(61%)	14/381(3%)

Query 16

IAEKLNLISIKDLNKAFSRKILREDEYKEIKTLLFKKEFKGIEKGTVIFLNDNLDVVRGYP

I E+ + L AF+R LR E ++ LLFKK +E GTVI+L D +V+RG+P

Sbjct 10

IQERTGVPSSRLKVAFAFGSLRLLESAGMQALLFKKPLGDLEAGTVIYLGDETEVIRGFP

69

Query 76

KTYRAITLYPTIKKHFIDKVVIEEKLNGYNIRIVKID-GEVYALTRSGYICPFTTKKVKK

K R + L PTI++HF D+V +EEK+NGYN+RI + GE ALTR G++CPFTT+K ++

Sbjct 70

KIRRTLLSPTIQEHFRDRVAVEEKMNGYNVRIACLSSGETVALTRGGHVCPTTRAAQ

129

Query 135

FLNT-FLDDYSEYMLCGEMIGTNNPVRVYKEDVDGENTGEVIEDIEKPEPNKSLPI

193

JP+

185

JPV

185

YT

253

YT

245

Sequences with E-value BETTER than threshold

☒ select all 4 sequences selected

PSI-BLAST iteration 1

	Description	Max score	Total score	Query cover	E value	Per. Ident	Accession	Select for PSI blast	Used to build PSSM	Newly added
☒	Chain A, Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]	313	313	95%	4e-104	42.26%	5D10_A	☒		
☒	Chain B, Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]	311	311	95%	3e-103	41.99%	5D10_B	☒		
☒	Chain A, PAB1020 [Pyrococcus abyssi GE5]	183	183	90%	2e-53	32.96%	2VUG_A	☒		
☒	Chain A, DNA ligase [Aquifex aeolicus]	158	158	85%	2e-44	30.59%	3QWU_A	☒		

Run PSI-BLAST Iteration 2 with max number of sequences

500

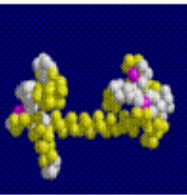
Run

Sequences with E-value WORSE than threshold

☒ select all 14 sequences selected

PSI-BLAST iteration 1

	Description	Max score	Total score	Query cover	E value	Per. Ident	Accession	Select for PSI blast	Used to build PSSM	Newly added
☒	Chain A, 6-phosphofructokinase [Geobacillus stearothermophilus]	31.6	31.6	31%	1.8	23.18%	4I4I_A	☐		
☒	Chain A, Crystal Structure Of A Phosphofructokinase Mutant From Bacillus Stearothermophilus Bound With Fructose-6-Phospl	30.8	30.8	31%	2.9	23.18%	1MTO_A	☐		
☒	Chain A, 4-cresol Dehydrogenase [hydroxylating] Flavoprotein Subunit [Pseudomonas putida]	30.8	30.8	8%	3.3	42.42%	1WVE_A	☐		
☒	Chain A, Crystal Structure Of P-Cresol Methylhydroxylase At 2.5 A Resolution [Pseudomonas putida]	30.8	30.8	8%	3.4	42.42%	1DIL_A	☐		
☒	Chain A, Trigger Factor [Vibrio cholerae]	30.8	30.8	24%	3.5	24.74%	1T11_A	☐		
☒	Chain A, 6-phosphofructokinase [Geobacillus stearothermophilus]	30.4	30.4	31%	3.8	23.18%	3U39_A	☐		
☒	Chain A, 6-phosphofructokinase [Geobacillus stearothermophilus]	30.4	30.4	31%	3.9	23.18%	4I36_A	☐		
☒	Chain A, 50S ribosomal protein L7Ae [Archaeoglobus fulgidus]	28.9	28.9	10%	5.0	34.88%	1RLG_A	☐		



Iteration 1

Descriptions

Graphic Summary

Alignments

Taxonomy

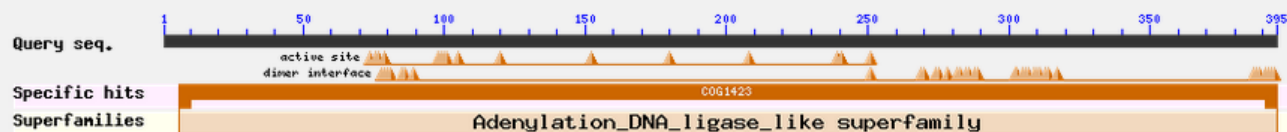
hover to see the title click to show alignments ☒ Show Conserved Domains

hover to see the title click to show alignments ☒ Show Conserved Domains

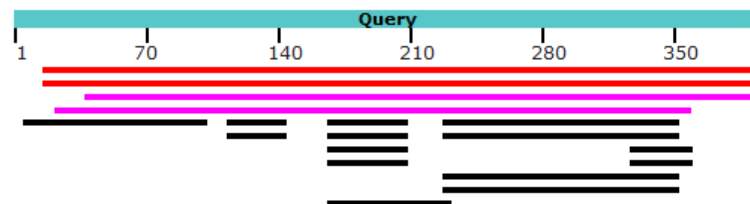
Alignment Scores < 40 40 - 50 50 - 80 80 - 200 >= 200

18 sequences selected ?

Putative conserved domains have been detected, click on the image below for detailed results.



Distribution of the top 18 Blast Hits on 18 subject sequences

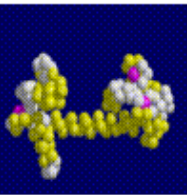


Run PSI-Blast iteration 2

Number of sequences

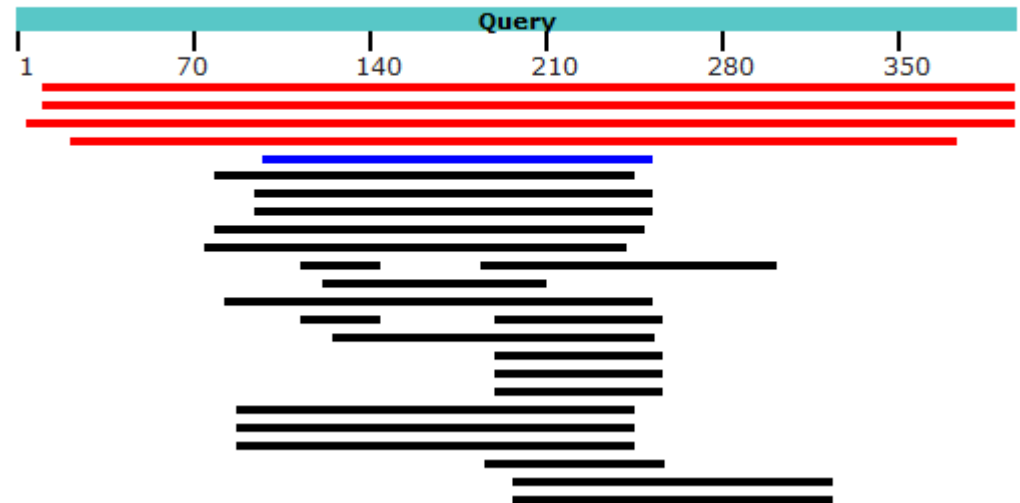
500

Run



Iteration 2

Distribution of the top 26 Blast Hits on 26 subject sequences



Old and new hits are indicated in the list

Sequences with E-value BETTER than threshold

☒ select all 5 sequences selected [Skip to the first new sequence](#)

PSI-BLAST iteration 2

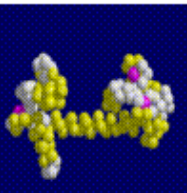
	Description	Max score	Total score	Query cover	E value	Per. Ident	Accession	Select for PSI blast	Used to build PSSM	Newly added
☒	Chain A, Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]	506	506	97%	5e-180	42.23%	5D1O_A	☒	☒	
☒	Chain B, Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]	504	504	97%	3e-179	41.97%	5D1O_B	☒	☒	
☒	Chain A, PAB1020 [Pyrococcus abyssi GE5]	442	442	98%	1e-154	31.73%	2VUG_A	☒	☒	
☒	Chain A, DNA ligase [Aquifex aeolicus]	425	425	88%	3e-148	29.38%	3QWU_A	☒	☒	
☒	Chain A, Human Dna Ligase Iii Recognizes Dna Ends By Dynamic Switching Between Two Dna Bound States [Homo sapiens]	47.0	47.0	38%	3e-05	19.28%	3L2P_A	☒		☒



Distribution of the top 29 Blast Hits on 29 subject sequences



Chain A, Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]	476	476	98%	4e-168	41.84%	5D1O_A	✓	✓
Chain B, Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]	474	474	98%	2e-167	41.58%	5D1O_B	✓	✓
Chain A, PAB1020 [Pyrococcus abyssi GE5]	442	442	98%	1e-154	31.73%	2VUG_A	✓	✓
Chain A, DNA ligase [Aquifex aeolicus]	409	409	88%	7e-142	29.38%	3QWU_A	✓	✓
Chain A, Human Dna Ligase Iii Recognizes Dna Ends By Dynamic Switching Between Two Dna Bound States [Homo sapiens]	187	187	38%	1e-53	19.28%	3L2P_A	✓	✓
Chain A, DNA ligase 4 [Homo sapiens]	56.6	56.6	39%	3e-08	18.64%	6BKF_A	✓	✓
Chain A, Dna Ligase 4 [Homo sapiens]	56.6	56.6	39%	3e-08	18.64%	3W1B_A	✓	✓
Chain A, Dna Ligase I [Homo sapiens]	55.9	55.9	72%	6e-08	18.54%	1X9N_A	✓	✓
Chain A, Thermostable Dna Ligase [Saccharolobus solfataricus]	55.1	55.1	39%	9e-08	16.37%	2HIV_A	✓	✓
Chain A, DNA ligase [Archaeoglobus fulgidus]	52.0	52.0	42%	7e-07	19.23%	3GDE_A	✓	✓
Chain A, Dna Ligase 4 [Homo sapiens]	42.8	42.8	28%	1e-04	20.00%	3VNN_A	✓	✓
Chain A, Dna Ligase [Thermococcus sibiricus MM 739]	45.1	45.1	40%	1e-04	17.82%	4EQ5_A	✓	✓
Chain A, Putative Dna Ligase-like Protein Rv0938/mt0965 [Mycobacterium tuberculosis H37Rv]	41.2	41.2	41%	0.001	22.02%	1VS0_A	✓	✓
Chain A, Dna Ligase [Thermococcus sp. 1519]	41.2	41.2	40%	0.002	14.94%	3RR5_A	✓	✓
Chain A, ATP-dependent DNA ligase [Mycobacterium tuberculosis]	40.5	40.5	41%	0.002	22.02%	6NHX_A	✓	✓
Chain A, ATP-dependent DNA ligase [Mycobacterium tuberculosis]	40.5	40.5	41%	0.002	22.02%	6NHZ_A	✓	✓



Significant 3D structure

Chain A, Atp-dependent Rna Ligase [Methanothermobacter thermautotrophicus str. Delta H]

Sequence ID: [5D10_A](#) Length: 378 Number of Matches: 1

Range 1: 1 to 378 [GenPept](#) [Graphics](#)

▼ [Next Match](#) ▲ [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
476 bits(1224)	4e-168	Composition-based stats.	164/392(42%)	238/392(60%)	16/392(4%)
Query 5	MMKVSAYDLNKKIAEKLNL	SIKDLNKA	FSRKILREDEYKEIKTLL	LFKKEFKGIEKGT	VIFL 64
Sbjct 1	MNSDIPFDL--IQERTGVP	SSRLKVA	FARGSLRLLESAGMQALL	FKKPLGLDLEAGT	VIYL 58
Query 65	NDNLDVVRGYPKTYRAIT	LYPTIKKH	FIDKVVIEEKLNGYNIR	IVKID-GEVYALTR	SGY 123
Sbjct 59	GDETEVIRGFPKIRRTL	LLSPTIQEH	FRDRVAVEEKMNGYNV	RIACLSSETVALTR	GGH 118
Query 124	ICPFTTRKKVKKFLNL-E	ILDDYSEY	MLCGEMIGINNPYTPY	YYYKEVDRGFENL	GFYIFDI 182
Sbjct 119	VCPFTTRKAQELLDLSE	FFREHPDL	VICGEMIGRDNPNVVS	QDYPEVG----PLG	FRVFDL 174
Query 183	KERETNKS	LPIKERIN	LCEKYNLPYVKPLAV	VDKDEAHIHVREIIE	KLNKEGREGVVLKD 242
Sbjct 175	REKNTNRPLPVEERRAL	LD	SYGLPNVRLFGVYPIE	EAASEVADIIRALG	MAGREGVVMKD 234
Query 243	PDMAVSPIKYTTHYTQ	CEDLKS	AFTFFFDLGMDFLFS	RVVREGFMSYEFK	ETLEERKNRA 302
Sbjct 235	PSMEVPPLKYTSSQA	HARELAYA	FSYPFDGFRPFFFS	RVIREGFQAYELDE	SDDETRERA 294
Query 303	KDLGEAILLPMVETIN	KVASGERV	SEDFELIFDSEEDF	DEFLDFMRKMKM	VITIKNIEKI 362
Sbjct 295	RRLGEAIIYPMLERIK	SISAGEA	AYEDTVIDVEDREA	AEFIRHLVRLGVS	SATLA----- 349
Query 363	DTEEGVKIKAVIGKI	YNKTND	KIISYLN	GTW 394	
Sbjct 350	DYRDG---RATIRRFY	QSTTD	RINNYL	KGGY 378	

Get the 3D structure ...

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PDB-101

WORLDWIDE
PDB
PROTEIN DATA BANK

EMDataResource
Unified Data Resource for 3DDEM

NUCLEIC ACID
DATABASE

Worldwide
Protein Data Bank
Foundation



Structure Summary

3D View

Annotations

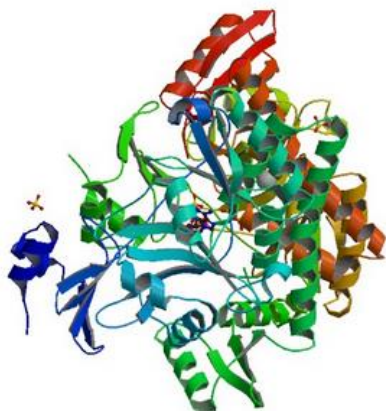
Sequence

Sequence Similarity

Structure Similarity

Experiment

Biological Assembly 1 ?



5D1O

Archaeal ATP-dependent RNA ligase - form 1

DOI: [10.2210/pdb5D1O/pdb](https://doi.org/10.2210/pdb5D1O/pdb)

Classification: [LIGASE](#)

Organism(s): [Methanothermobacter thermautotrophicus \(strain ATCC 29096 / DSM 1053 / JCM 10044 / NBRC 100330 / Delta H\)](#)

Expression System: [Escherichia coli](#)

Deposited: 2015-08-04 Released: 2016-03-09

Deposition Author(s): [Murakami, K.S.](#)

Funding Organization(s): National Institutes of Health/National Institute of General Medical Sciences

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 2.648 Å

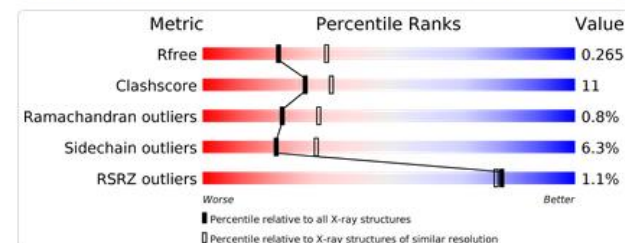
R-Value Free: 0.265

R-Value Work: 0.206

wwPDB Validation

3D Report

Full Report



3D View: [Structure](#) | [Electron Density](#) | [Ligand Interaction](#)

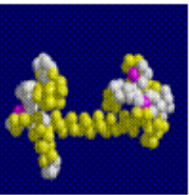
Standalone Viewers

[Protein Workshop](#) | [Ligand Explorer](#)

Global Symmetry: Cyclic - C2 ⓘ (3D View)

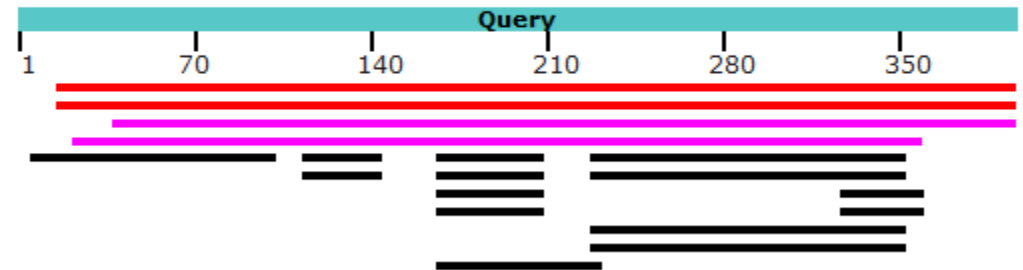
Global Stoichiometry: Homo 2-mer - A2 ⓘ

This is version 1.2 of the entry. See complete [history](#).

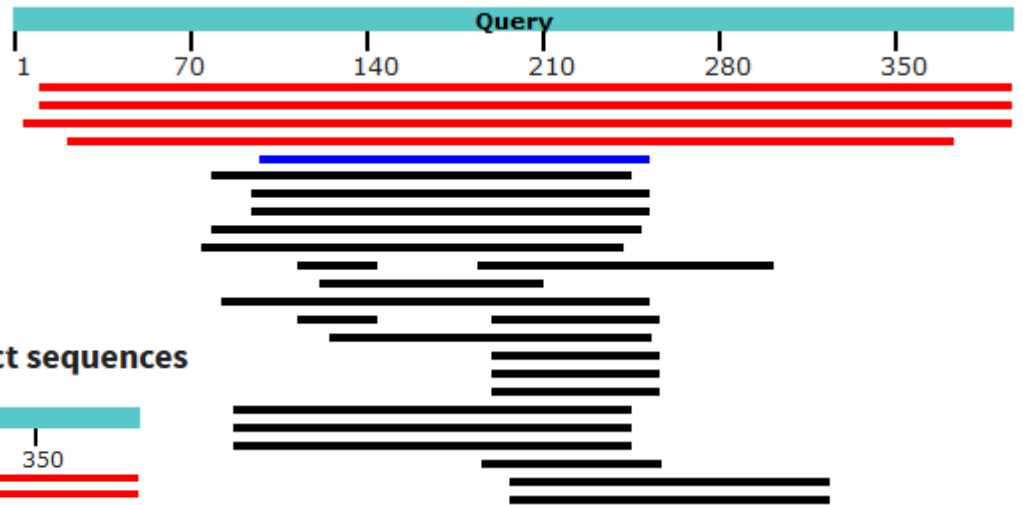


The search process

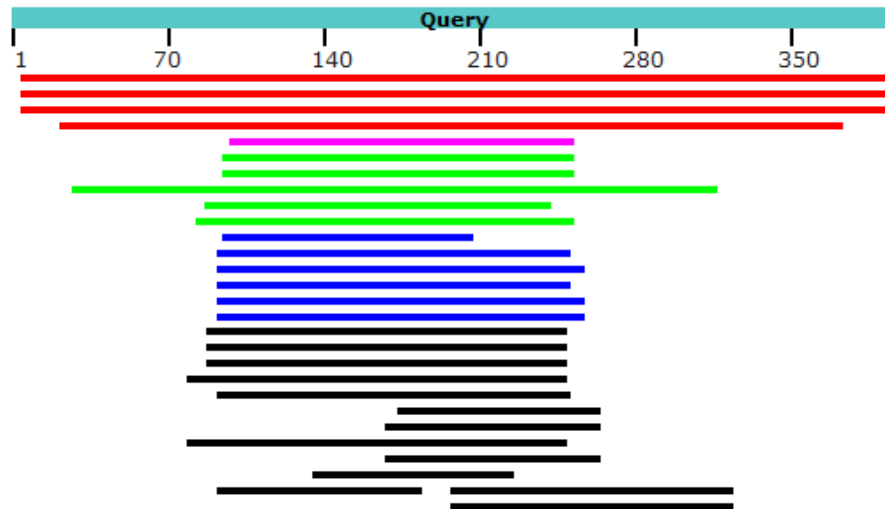
Distribution of the top 18 Blast Hits on 18 subject sequences

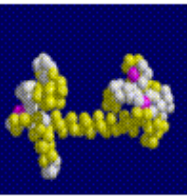


Distribution of the top 26 Blast Hits on 26 subject sequences



Distribution of the top 29 Blast Hits on 29 subject sequences





What is the function of ExTox?

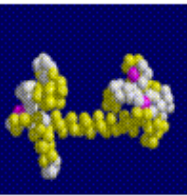
- ❑ PSI-Blast searching identifies a trypsin as similar
- ❑ We know which trypsin residues that are essential for activity
- ❑ These residues are conserved in ExTox
- ❑ ExTox may therefore have trypsin-like activity!

```

      ↓
ExTox  : GQTSATGVLIGKNTVLTNRHIAKFANGDPSKVSFRPSINTDDNGNTETPYGEYEVKEILQ
          G      G L+ +N V++ H K          +V          G+ +          ++
Trypsin: GYHFCGGS�VNENWVVSAAHCYKSR---VEVRLGEHNIKVTEGSEQFISSSRVIRHPNY
          ↓
ExTox  : EPFGAGVDLALIRLKPDQNGVSLGDKISPAKIGTSNDLKDGDKLELIGYPFDHKVNQMHR
          +      D+ LI+L          +L      + P      + TS      G      + G+      + ++
Trypsin: SSYNIDNDIMLIKLS---KPATLNTYVQPVALPTS-CAPAGTMCTVSGW--GNTMSSTAD
          102
ExTox  : SE-----IELTT-----LSRGLRYYGFTVP-----GNSGSGIFNSNGELVGI
          S          L+          ++ +      G+          G+SG +      NGEL G+
Trypsin: SNKLQCLNIPILSYSDCNNSYPGMITNAMFCAGYLEGGKDSCQGDSSGGPVVC-NGELQGV
          195
ExTox  : HSSKVSHLDREHQINYGV--GIGNYV
          S          + +      Y          +++
Trypsin: VSWGYGCAEPGNPGVYAKVCIFNDWL

```

TIBS



What about related proteins?

- ❑ This is a PSI-Blast search with ExTox (Query)
- ❑ Can this hit (Sbjct) have similar activity?
 - The sequence identity is very low (17%)

[Download](#) ▼
 [GenPept](#)
[Graphics](#)
 ✓ -used in PSSM

Chain A, Crystal Structure Of The Degrading Stress Sensor [Escherichia coli]

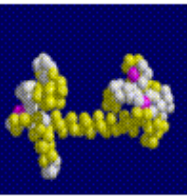
Sequence ID: [1SOT_A](#) Length: 320 Number of Matches: 1

[See 5 more title\(s\)](#) ▼
 [See all Identical Proteins\(IPG\)](#)

Range 1: 32 to 197 [GenPept](#) [Graphics](#)

▼ [Next Match](#) ▲ [Previous Match](#)

	Score	Expect	Method	Identities	Positives	Gaps	
	119 bits(299)	2e-31	Composition-based stats.	32/191(17%)	67/191(35%)	36/191(18%)	
Query	88	FVKGQTSATGVLLIG-KNTVLTNRHIAKFANGDPSKVSFRPSINTDDNGNTETPYGEYEVK					146
		++ +T +GV+ + ++TN+H+ A+ + +E					
Sbjct	32	QLEIRTLGSGVIXDQRGYIITNKHVINDADQIIIVALQD-----GRVFEAL					76
Query	147	EILQEPFGAGVDLALIRLKPQNGVSLGDKISPAKIGTSNDLKDGDKLELIGYPFD----					202
		+ + + DLA++++ + + I GD + IG P++					
Sbjct	77	LVGSD---SLTDLAVLKI-----NATGGLPTIPINARRVPHIGDVVLAIGNPYNLGQT					126
Query	203	---HKVNQMHRSEIELTTLRGLRYYGFTVPGNSGSGIFNSNGELVGIHSSKVSH---LD					256
		++ R + T L+ GNSG + NS GEL GI++ +					
Sbjct	127	ITQGIISATGRIGLNPTGRQNFLOTDA SINHGNSGGALVNSLGELXGINTLSFDKSN DGE					186
Query	257	REHQINYGVGI 267					
		I ++					
Sbjct	187	TPEGIGFAIPF 197					



Comparison of the alignments

Download ▾ GenPept Graphics ✓ -used in PSSM

Chain A, Crystal Structure Of The Degrading Stress Sensor [Escherichia coli]

Sequence ID: [1SOT_A](#) Length: 320 Number of Matches: 1

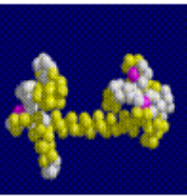
[See 5 more title\(s\)](#) ▾ [See all Identical Proteins\(IPG\)](#)

Range 1: 32 to 197 [GenPept](#) [Graphics](#)

▾ [Next Match](#) ▲ [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
119 bits(299)	2e-31	Composition-based stats.	32/191(17%)	67/191(35%)	36/191(18%)
Query 88	FVKGQTSATGVLIG-KNTVLTNRHIAKFANGDPSKVSFRPSINTDDNGNTETPYGEYEVK	146			
	++ +T +GV+ + ++TN+H+ A+ + +E				
Sbjct 32	QLEIRTLGSGVIXDQRGYITITNKHVINADQIIIVALQD-----GRVFEAL	76			
Query 147	EILQEPFGAGVDLALIRLKPDPQNGVSLGDKISPAKIGTSNDLKDGDKLELIGYPFD----	202			
	+ + + DLA++++ + + I GD + IG P++				
Sbjct 77	LVGSD---SLTDLAVLKI-----NATGGLPTIPINARRVPHIGDVVLAIGNPYNLGQT	126			
Query 203	---HKVNQMHRSEIELTTLRGLRYYGFTYPGNSGSGIFNSNGELVGIHSSKVSCH---LD	256			
	++ R + T L+ GNSG + NS GEL GI++ +				
Sbjct 127	ITQGIISATGRIGLNPTGRQNFLQTDASINHGNSGGALVNSLGELXGINTLSFDKSNDE	186			
Query 257	REHQINYGVGI	267			
	I + +				
Sbjct 187	TPEGIGFAIPF	197			

- The same is the case for D and S
- It is possible that Degrading Stress Sensor (Sbjct) has trypsin-like activity!
 - *Must be confirmed experimentally ...*



Comparison of the alignments

NRHIA

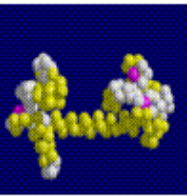
ExTox : GQTSATGVLIGKNTVLTNRHIAKFANGDPSKVSFRPSINTDDNGNTETPYGEYEVKEILQ
 G G L+ +N V++ H K +V G+ + ++
 Trypsin: GYHFCGGSLVNENWVVSAAHICYKSR---VEVRLGEHNIKVTEGSEQFISSSRVIRHPNY

57

NRHIA

Query 88 FVKGQTSATGVLIG-KNTVLTNRHIAKFANGDPSKVSFRPSINTDDNGNTETPYGEYEVK 146
 ++ +T +GV+ + ++TN+H+ A+ + +E
 Sbjct 32 QLEIRTLGSGVIXDQRGYIITNKHVINDADQIIVALQD-----GRVFEAL 76

- The numbering of residues is different
- Looking at the sequence (NRHIA) shows us that the H is conserved



Useful web sites

- EBI

- <http://www.ebi.ac.uk/>

- NCBI

- <http://www.ncbi.nlm.nih.gov/>

- ExPASy

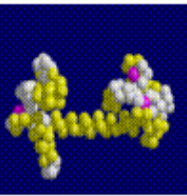
- <http://www.expasy.org/>

- ELIXIR

- <https://www.elixir-europe.org/>

- ELIXIR Norway

- <https://www.elixir-norway.org/>



Bioinformatics

“Bioinformatics can be defined as the development and use of computational and mathematical methods for the acquisition, archiving, analysis and interpretation of biological information to determine biological functions and mechanisms as well as their application in user communities.”

Biotechnology and Biological
Sciences Research Council

<http://www.bbsrc.ac.uk>



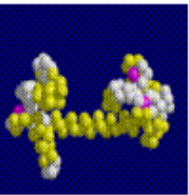
Scientific American, March 2001



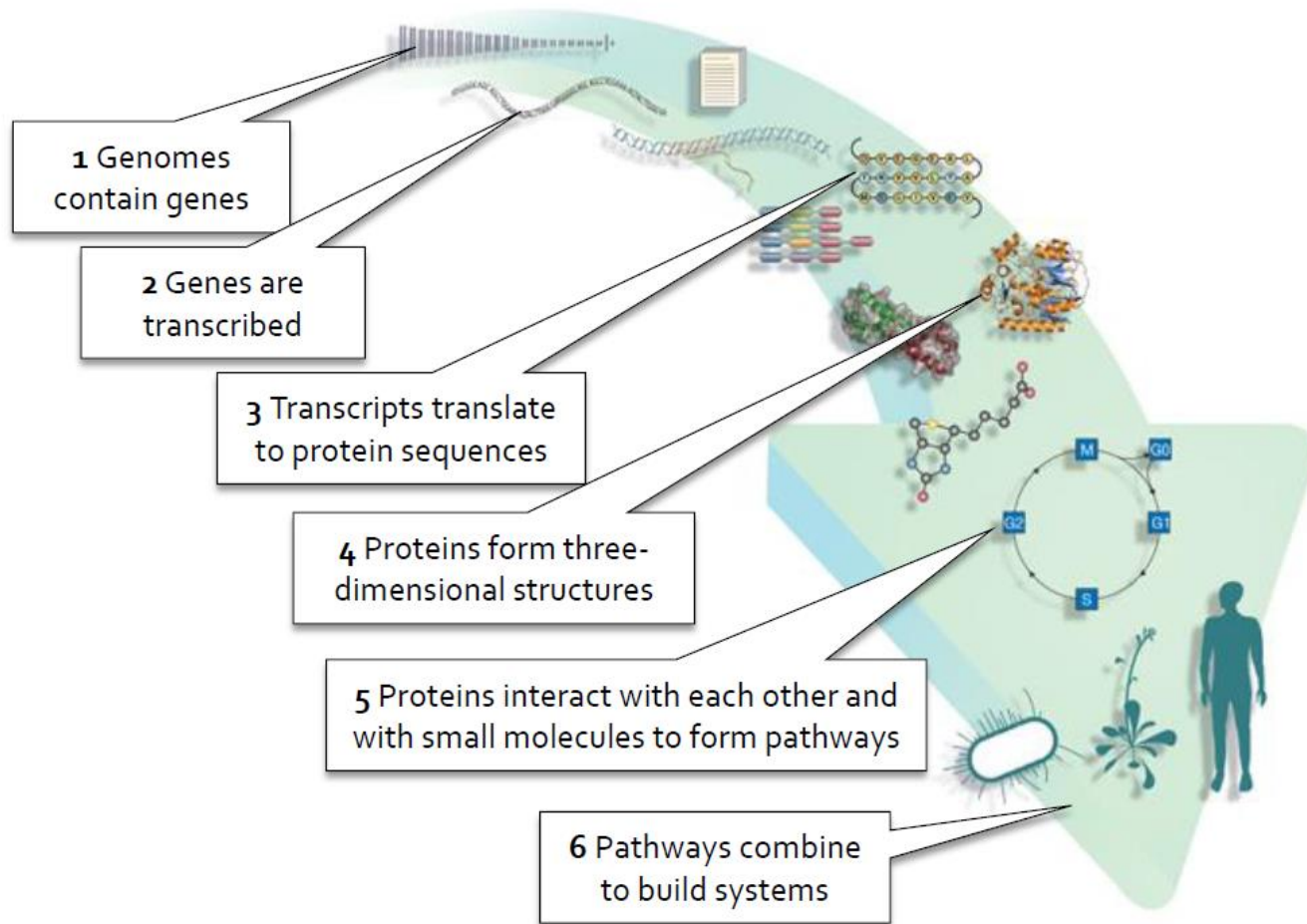
Bioinformatics

“In biology, Bioinformatics */ˌbaɪ.ɒʊˌɪnfərˈmætɪks/* is an interdisciplinary field that develops and improves upon methods for storing, retrieving, organizing and analyzing biological data.”

– Wikipedia



Bioinformatics underpins life-science research



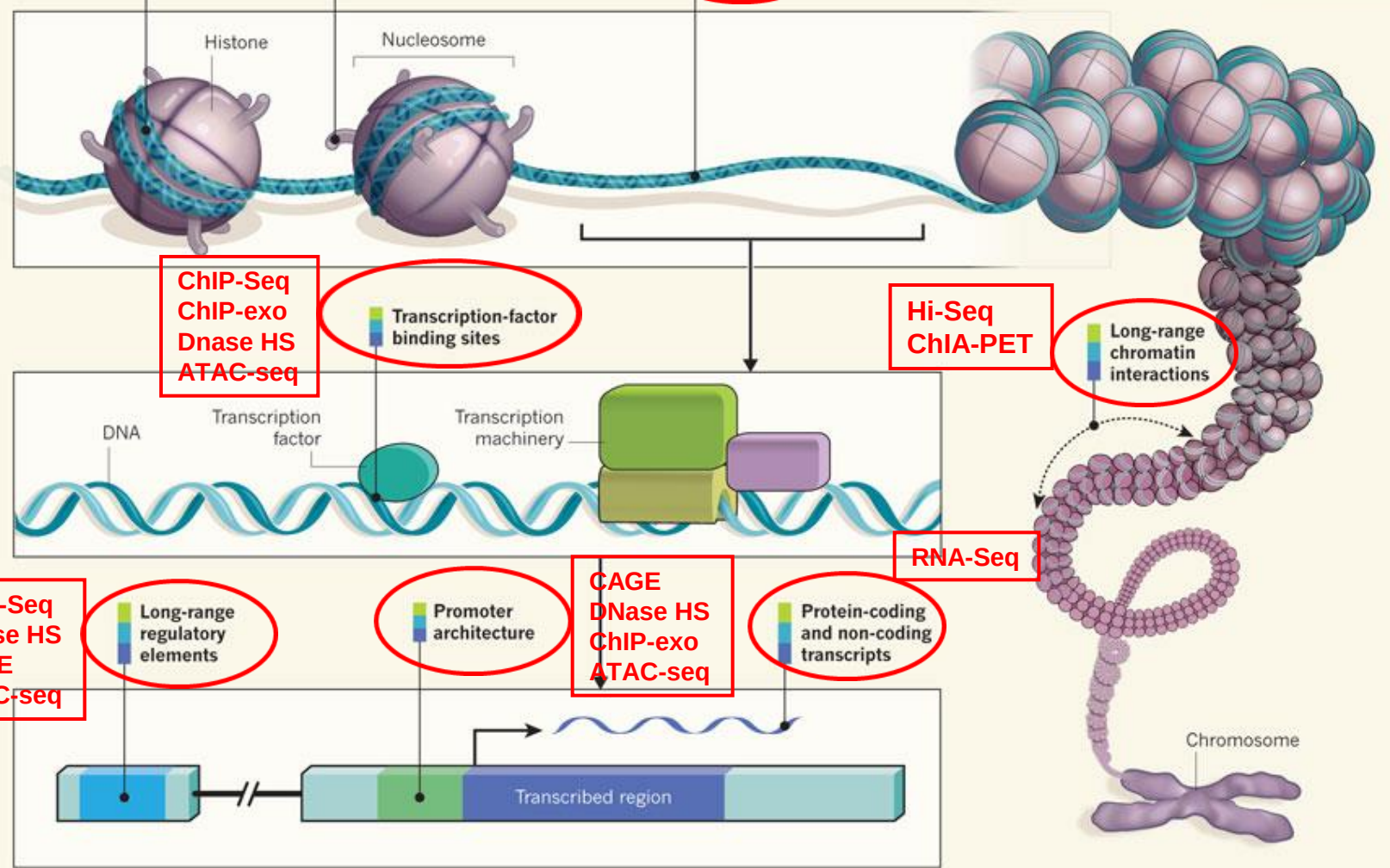


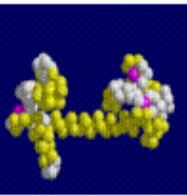
RRBS/WGSBS

ChIP-Seq

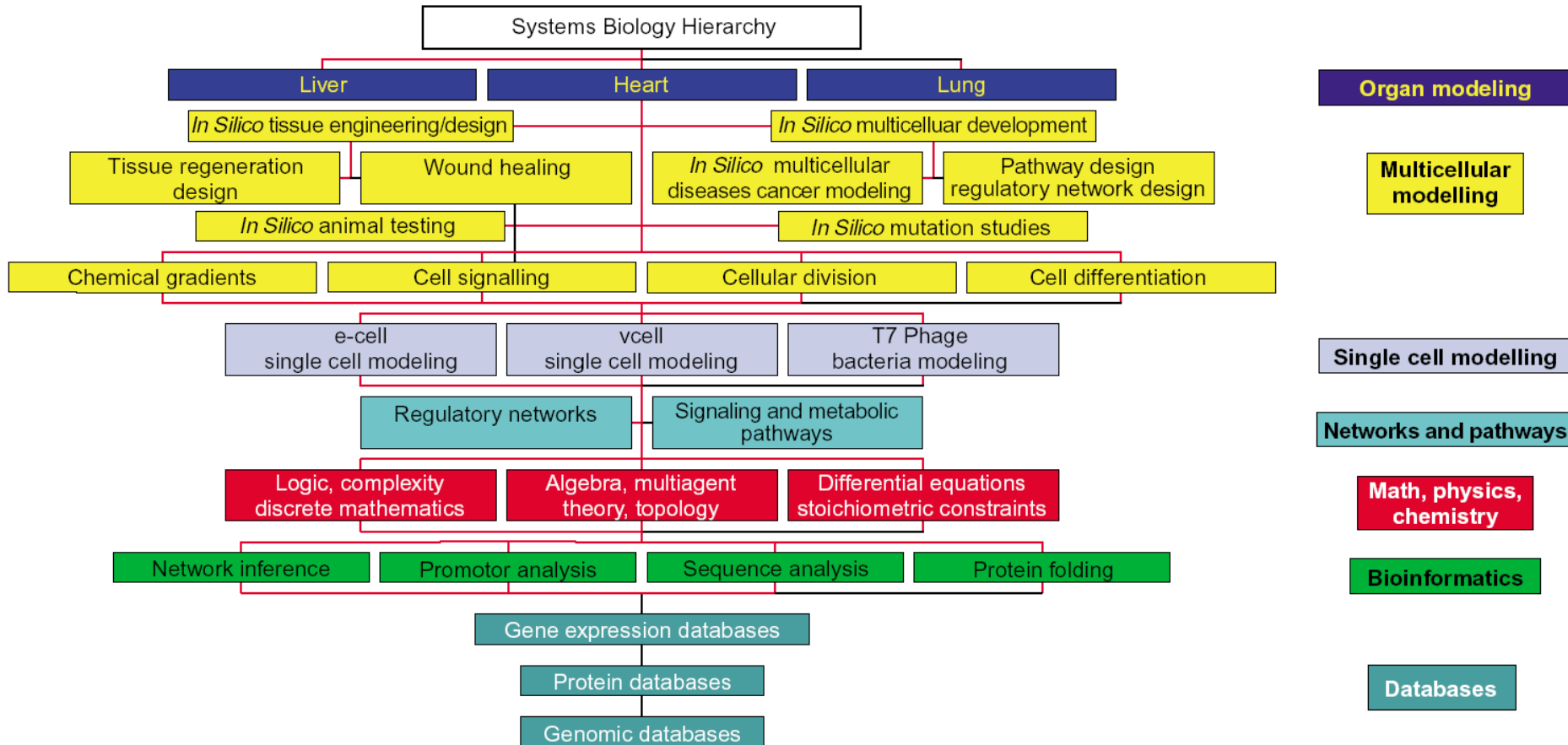
Open Chromatin

Functional genomic elements

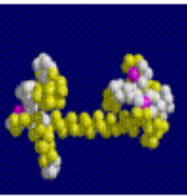




Systems biology hierarchy



genes → RNA → proteins → interactions → complexes → networks →
cells → tissues → organisms → populations → ecosystems



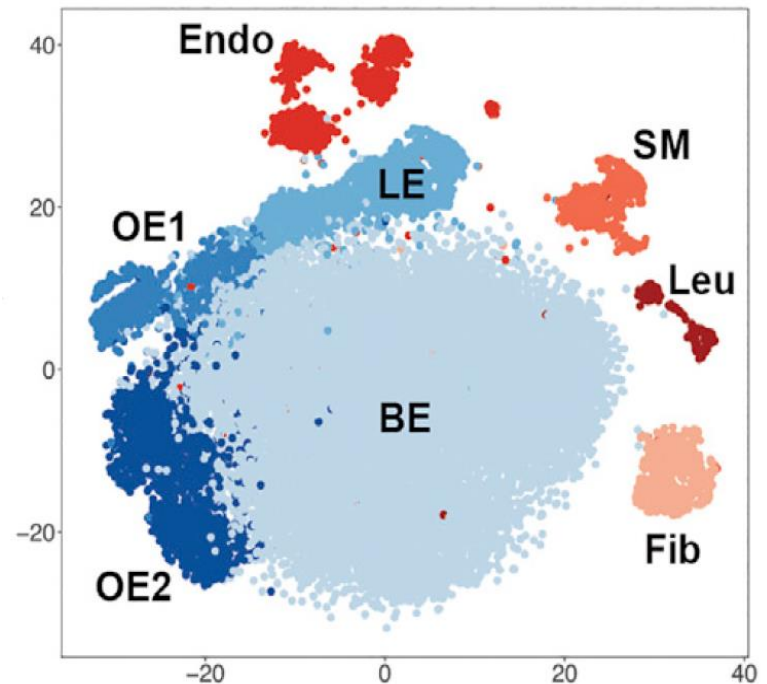
Single-cell Transcriptomics

- ❑ RNA-Seq require less amount of starting material.
- ❑ Sequencing of RNA-levels representing single-cell quantities now possible (~10pg)
- ❑ Methods for selecting and sequencing single cells have recently are currently in progress
- ❑ **Applications:**
 - Stratification of expression from various cell-types in a heterogeneous mixture or tissue
 - Transcription for rare cell entities in poluations, like Circulating Tumor Cells
 - Heterogeneity in cell-transcription of cells in a population as determinants of cell fate and differentiation
 - Pseudo «time-series» from a single sample

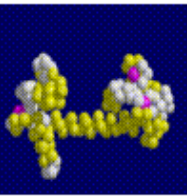
Cell Reports

Resource

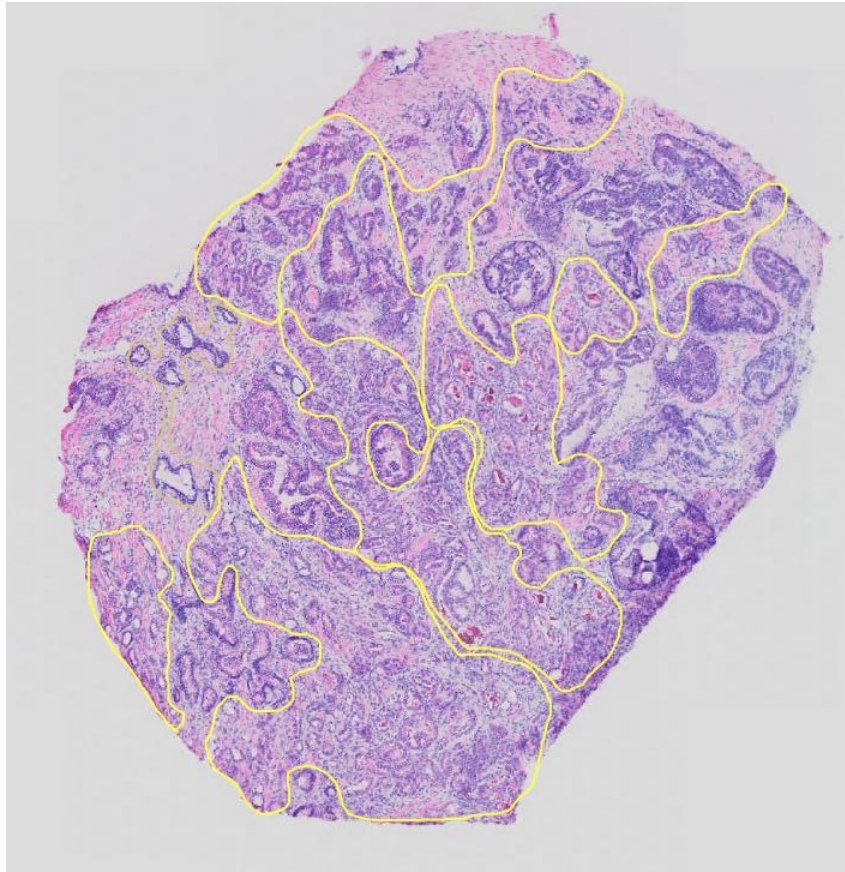
A Cellular Anatomy of the Normal Adult Human Prostate and Prostatic Urethra



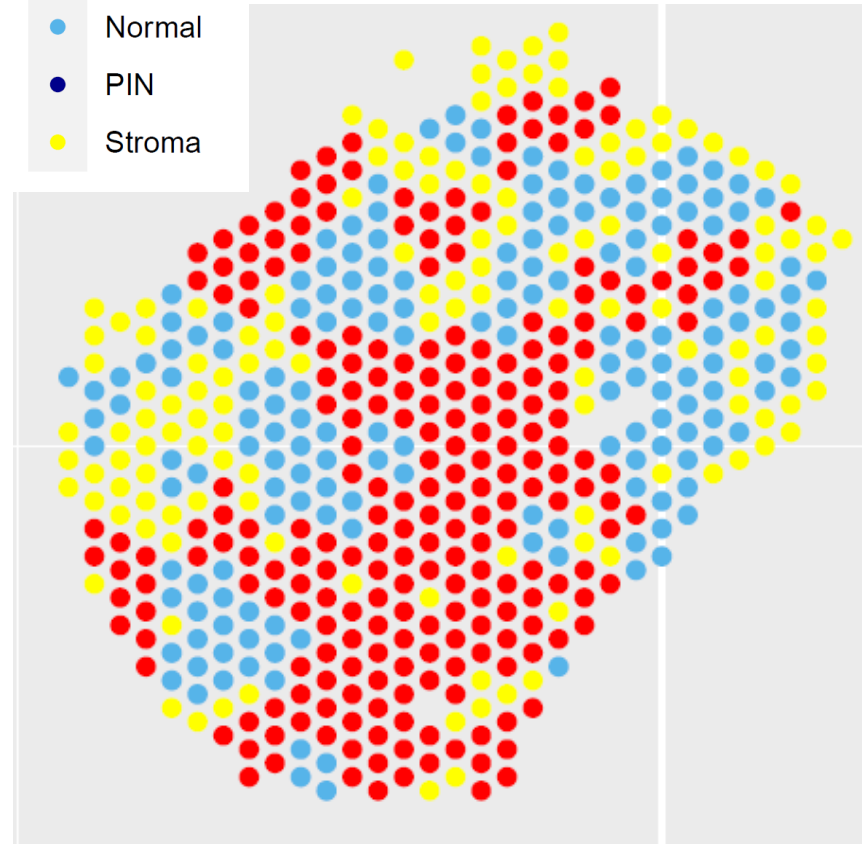
Gervaise et al., 2018

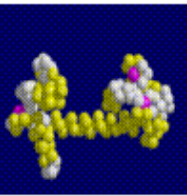


Spatial Transcriptomics



- Cancer
- Inflamm
- Normal
- PIN
- Stroma

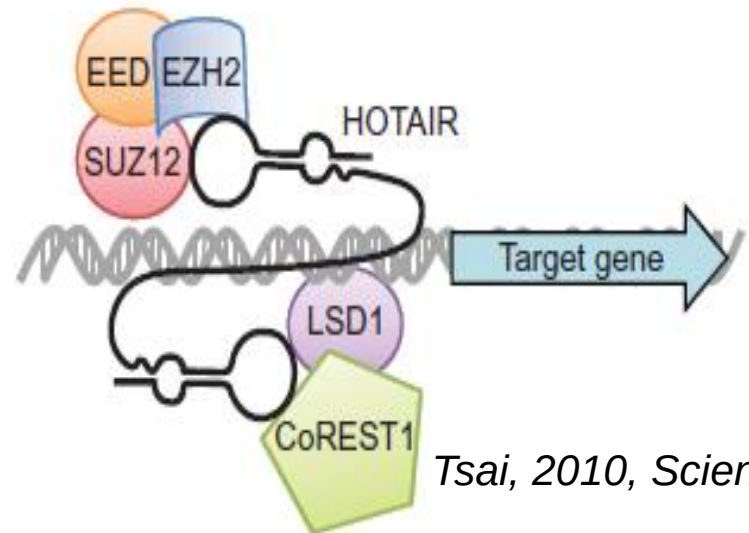




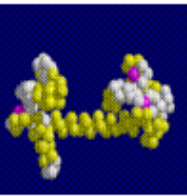
Long non-coding RNA (lncRNA)

- ❑ Not translated into protein (though it has been shown that many translate short reading frames into peptides)
- ❑ Current estimates suggest about 20-40 000 lncRNAs in human.
- ❑ More cell-type specific expression patterns compared to coding genes
- ❑ Many important function in the cell
 - *Impact comparable to protein coding genes*
 - *Affect epigenetics and gene regulation*
- ❑ Many other classes of short and long ncRNA,

Flexible scaffold for chromatin-modifying complexes



Tsai, 2010, Science

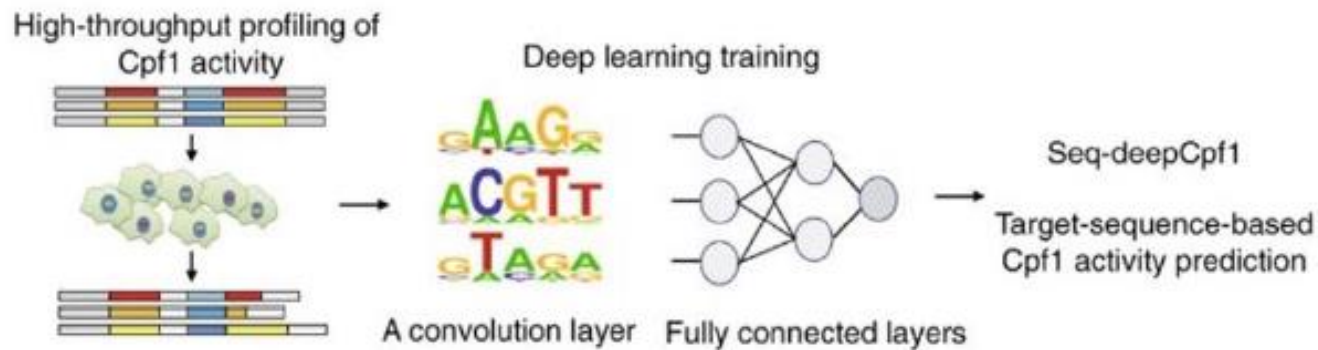


Small RNA analysis

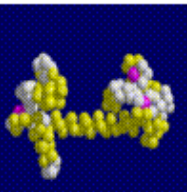
miRNA target prediction

siRNA design

gRNA (guide-RNA) design for CRISPR analysis

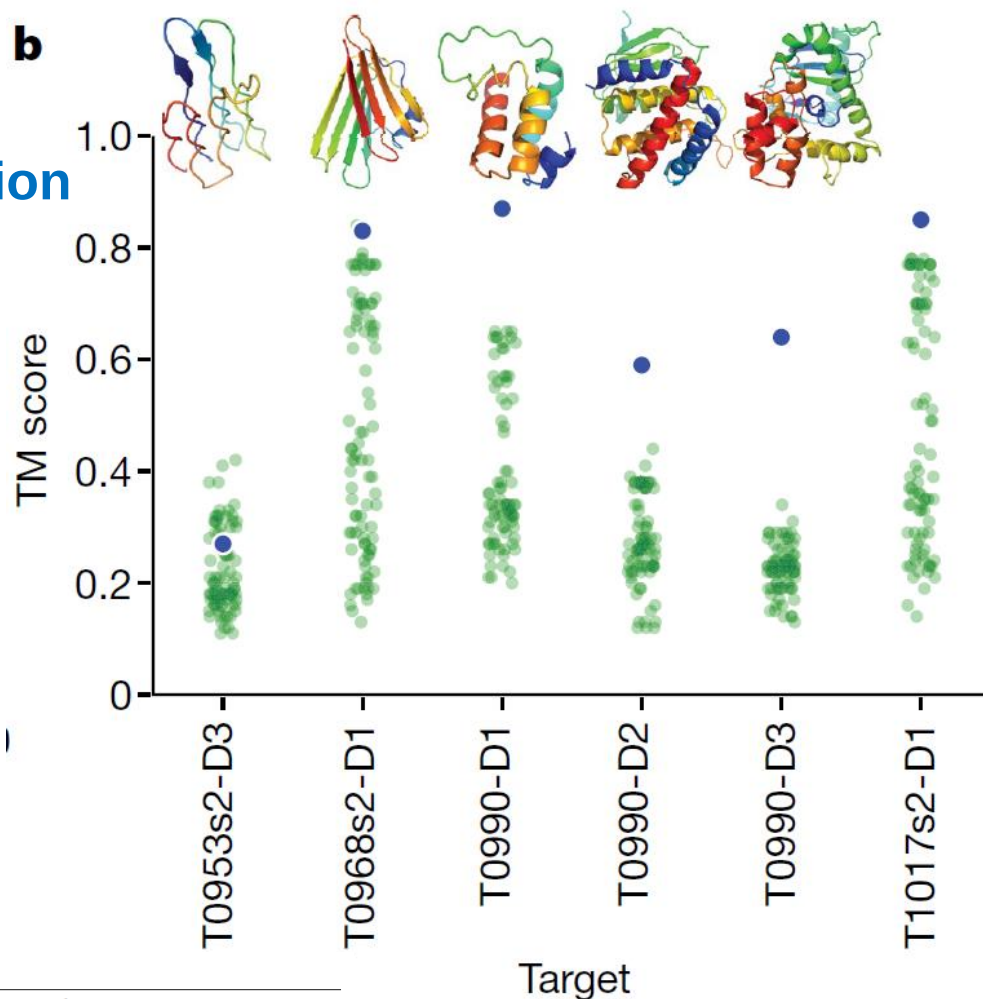


Kim et al., Nature Biotechnology, 2018



AlphaFold – AI for Protein Structure Prediction

- AlphaFold show that Free Modelling (without using templates) can match performance of Template Based Modelling, and reach the accuracy needed to provide biological insight.



Article

Improved protein structure prediction using potentials from deep learning

<https://doi.org/10.1038/s41586-019-1923-7>

Received: 2 April 2019

Accepted: 10 December 2019

Published online: 15 January 2020

Andrew W. Senior^{1,4*}, Richard Evans^{1,4}, John Jumper^{1,4}, James Kirkpatrick^{1,4}, Laurent Sifre^{1,4}, Tim Green¹, Chongli Qin¹, Augustin Židek¹, Alexander W. R. Nelson¹, Alex Bridgland¹, Hugo Penedones¹, Stig Petersen¹, Karen Simonyan¹, Steve Crossan¹, Pushmeet Kohli¹, David T. Jones^{2,3}, David Silver¹, Koray Kavukcuoglu¹ & Demis Hassabis¹

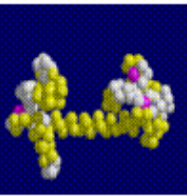


THE CANCER GENOME ATLAS

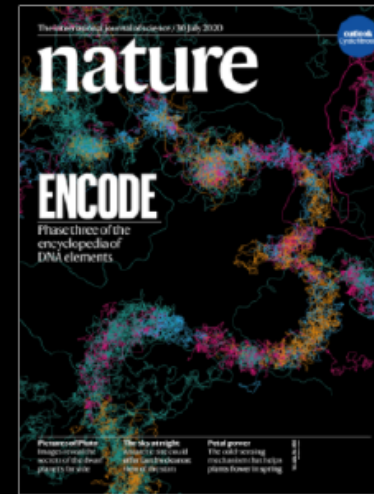
National Cancer Institute
National Human Genome Research Institute

Available Cancer Types	# Cases Shipped by BCR*	# Cases with Data*	Date Last Updated (mm/dd/yy)
Acute Myeloid Leukemia [LAML]	200	200	12/14/15
Adrenocortical carcinoma [ACC]	80	80	01/25/16
Bladder Urothelial Carcinoma [BLCA]	412	412	02/01/16
Brain Lower Grade Glioma [LGG]	516	516	01/25/16
Breast invasive carcinoma [BRCA]	1100	1097	01/25/16
Cervical squamous cell carcinoma and endocervical adenocarcinoma [CESC]	308	307	01/25/16
Cholangiocarcinoma [CHOL]	36	36	01/25/16
Colon adenocarcinoma [COAD]	461	461	02/01/16
Esophageal carcinoma [ESCA]	185	185	01/25/16
FFPE Pilot Phase II [FPPP]	38	38	01/25/16
Glioblastoma multiforme [GBM]	529	528	02/02/16
Head and Neck squamous cell carcinoma [HNSC]	528	528	01/25/16
Kidney Chromophobe [KICH]	66	66	01/25/16
Kidney renal clear cell carcinoma [KIRC]	530	530	01/25/16
Kidney renal papillary cell carcinoma [KIRP]	291	291	01/25/16
Liver hepatocellular carcinoma [LIHC]	377	377	01/25/16
Lung adenocarcinoma [LUAD]	521	521	01/25/16
Lung squamous cell carcinoma [LUSC]	510	504	01/25/16
Lymphoid Neoplasm Diffuse Large B-cell Lymphoma [DLBC]	48	48	01/25/16

- **Tissue samples from over 30 cancer and cancer subtypes**
- Exome sequencing
- Whole Genome Sequencing
- RNA-Seq
- Copy Number Variation (SNP)
- miRNA
- DNA Methylation
- Proteomic arrays (RPPA)
- Clinical data



2020: ENCODE 3



The print edition of *Nature*, 30 July 2020.

2020 marks the year of the publication of the third phase of the ENCODE project, which has been further expanded in terms of assays used, and cells and tissues analysed.

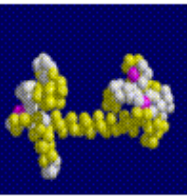
The [Encyclopedia paper of ENCODE 3](#), published in *Nature*, gives an overview of the various assays that were performed in human and mouse cell lines and tissues and describes a Registry of human and mouse candidate *cis*-regulatory elements (cCREs).

A suite of companion papers gives additional in-depth insights into the new datasets that were generated, their application to questions about how the activity of human and mouse functional elements is regulated, the organization of elements in the genome and nucleus, and the computational tools that were developed.

Nature

Expanded encyclopaedias of DNA elements in the human and mouse genomes.

The ENCODE Project Consortium *et al.*



Human microbiome



NIH HUMAN
MICROBIOME
PROJECT

Current News

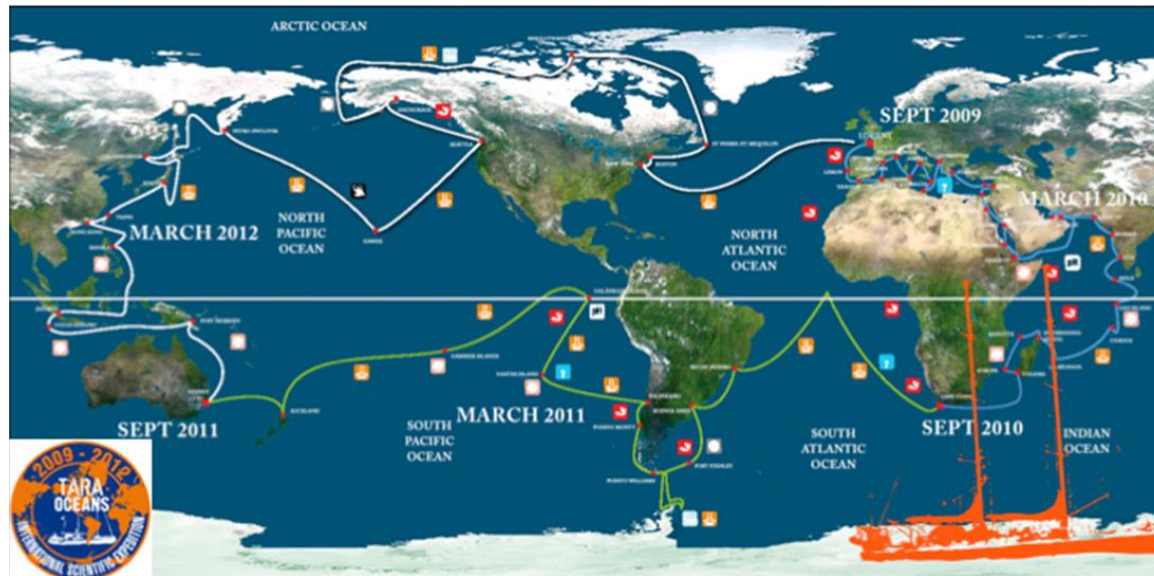
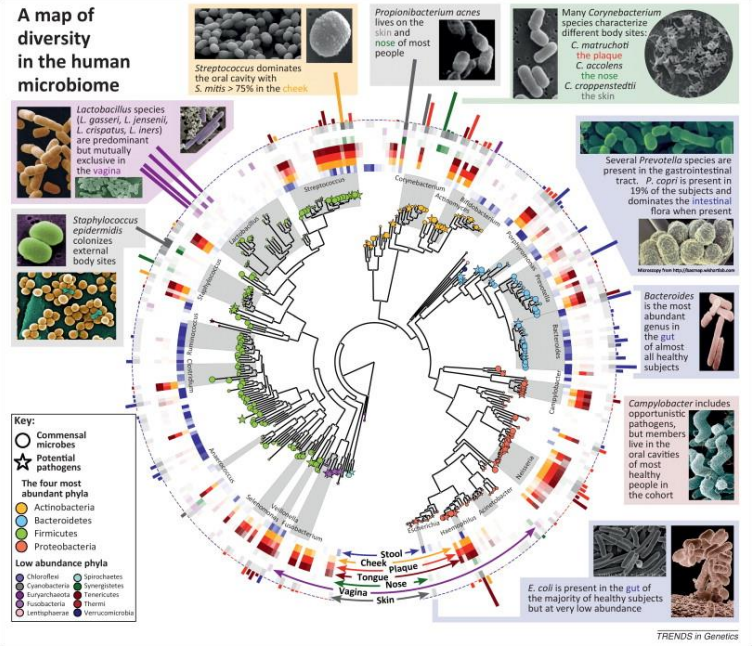
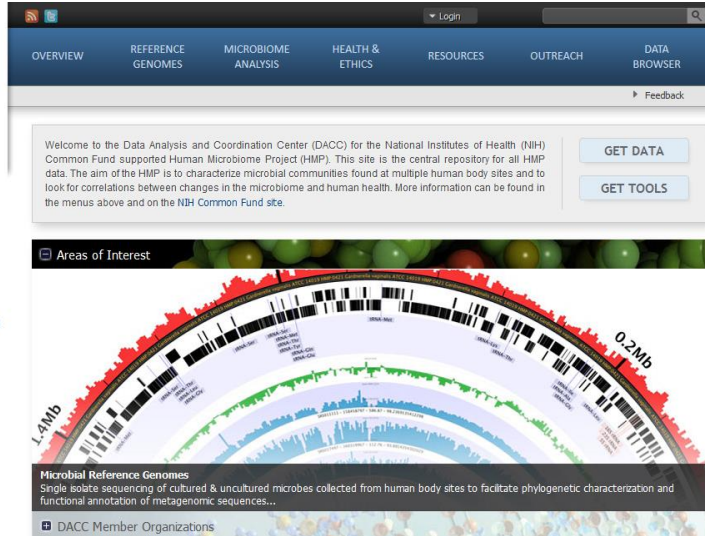
- June 2016
Poster and Booth at ASM 2016
- January 2015
Metagenome Analysis Workshop
March 3-6
- September 2014
IHMC 2015 from Mar. 31 to Apr. 2

[More News Items](#)

Publications

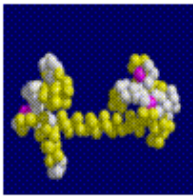
- A New Catalog of Microbiological
Tools for Women's Infectious
Disease...
- Genome Sequences of Nine
Gram-Negative Vaginal Bacterial
Isolates...
- Genome Sequences of 14
Firmicutes Strains Isolated from
the Human Vag...

[More Publications](#)



Sampling the oceans





Learning bioinformatics

- MOL3021/3022 Applied Bioinformatics
 - Intensive master level course, every spring
 - Focus on applied bioinformatics, with project